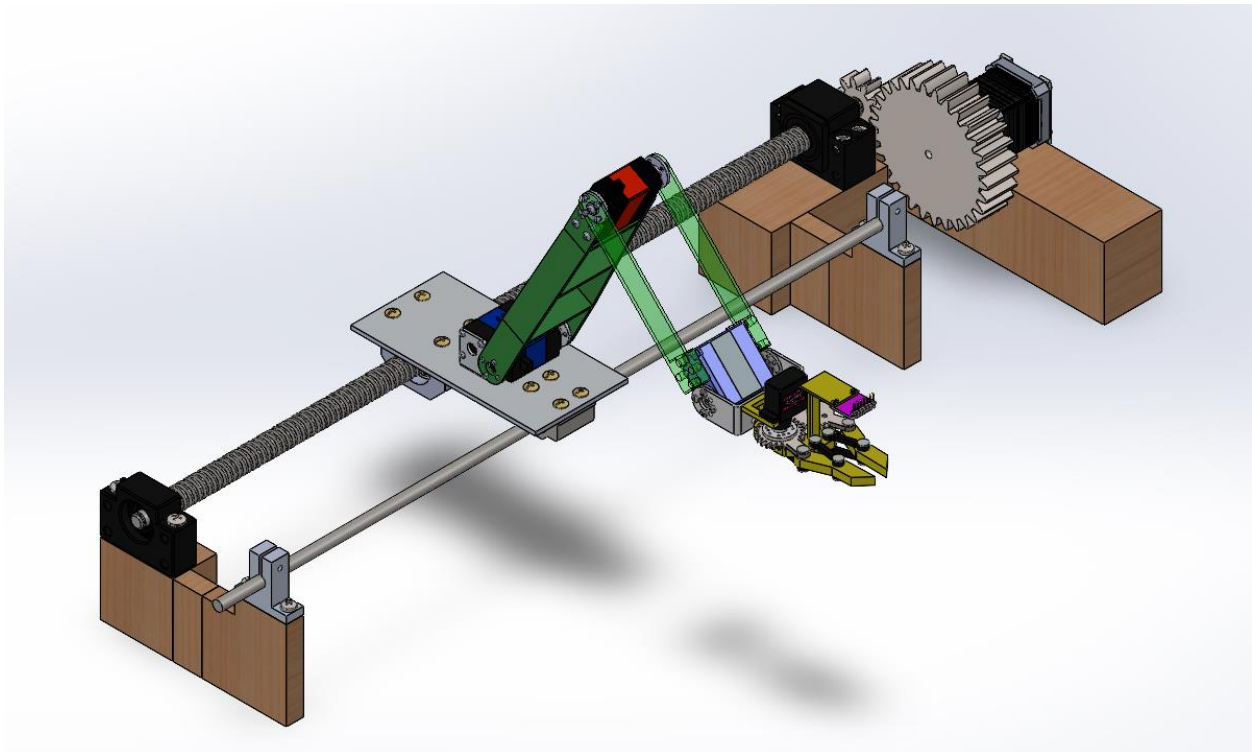


FDR Summary Package

Team 4

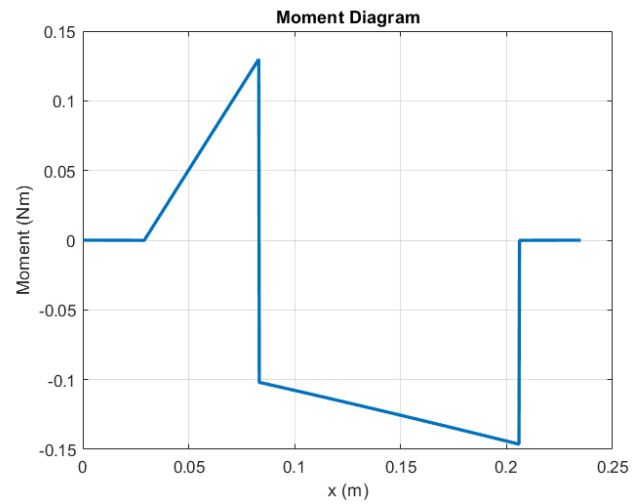
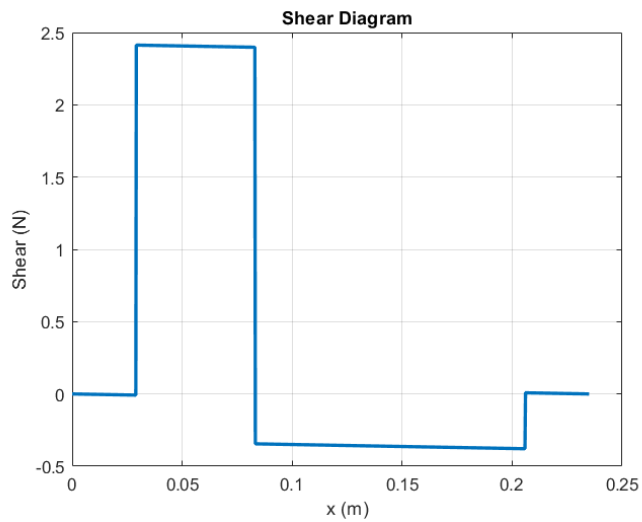
Members: Hunter Bellina, Ryan Brancheau, Rohan Kelkar, Pedro Pablo Labra Escudero, Tyler Seward, Kyle VanHorn, Mikey Wiesecke



Base

Shear-Moment Equations:

Shear:



Distance (m)	Equation
0 – 0.0290	$-0.277x$
0.0290 – 0.0832	$-0.277x + 2.421$
0.0832 - 0.206	$-0.277x - 0.322$
0.2060 – 0.2350	$-0.277x + 0.065$

Moment:

Distance (m)	Equation
0 – 0.0290	$-0.113x^2$
0.0290 – 0.0832	$-0.113x^2 + 2.421x - 0.070$
0.0832 - 0.206	$-0.113x^2 - 0.322x - 0.074$
0.2060 – 0.2350	$-0.113x^2 + 0.065x - 0.008$

Bending Stress

Variable	Description	Value
M	Max moment	-0.1487 Ncm
c	Distance from COG	0.15 cm
I	Moment of inertia	6.2678 x 10 ⁻¹⁰

bendstress =

0.3766

(units are MPa) $\sigma = \frac{Mc}{I}$

Shear Stress

Variable	Description	Value
V	Max Shear stress	2.43 N
Q	Area times distance from centroid	0.0007 m ²
I	Moment of inertia	6.2678 x 10 ⁻¹⁰ m ⁴
b	Width from neutral axis	0.0325 m

$$\tau = \frac{VQ}{Ib}$$

shear =

0.0975

units in MPa

Safety Factor

Variable	Value
Von Mises Stress	0.3766 MPa
Yield strength (Al 6061)	276 MPa

$$SF = \frac{\text{Yield Strength}}{\text{Von Mises Stress}}$$

sf =

732.8282

Deflection

Variable	Description	Value
W	Distributed load	4.08 N/m
P	Point load	2.744 N
d	right end of beam to right pin	0.021 m
E	Modulus of elasticity (6061 Al)	69 GPa
l	Distance between pins	0.185 m
L	Total length of beam	0.235 m
c	Distance from left end of beam to left pin	0.029 m
a	Distance from left pin to point load	0.054 m
b	Distance from right pin to point load	0.062 m

$$\delta = \frac{Wx(l-x)}{24EIL} * (x(l-x) + l^2 - 2(d^2 + c^2) - \left(\frac{2}{l}\right) * (d^2x + c^2(1-x))) + \frac{Wbx}{6EI} * (d^2 - x^2 - b^2)$$

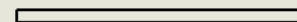
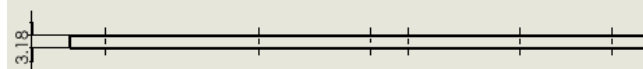
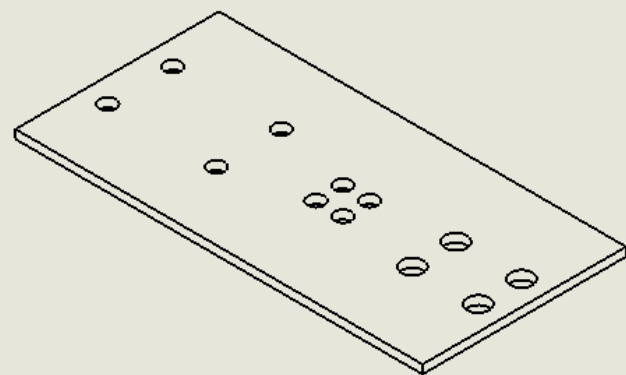
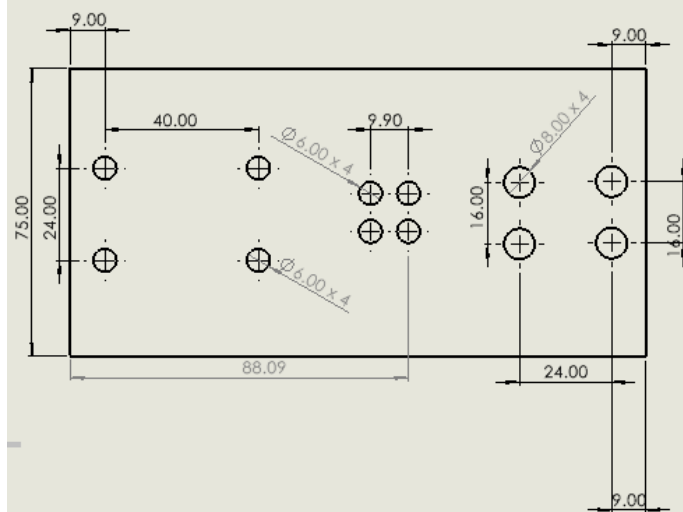
Equation derived from Engineering Handbook using superposition of Case 7 point deflection and Case 3 point deflection. Engineering Handbook stated there was no clean way to find deflection at critical points for Case 7.

deflect =

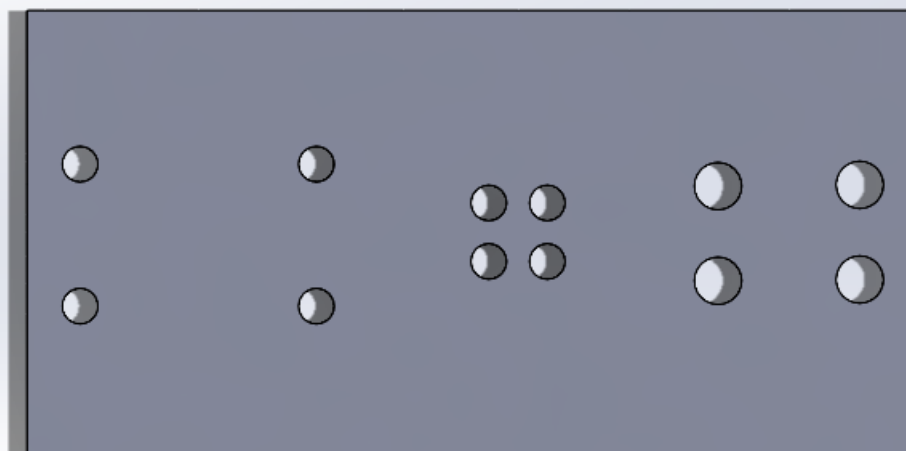
1.2168e-18

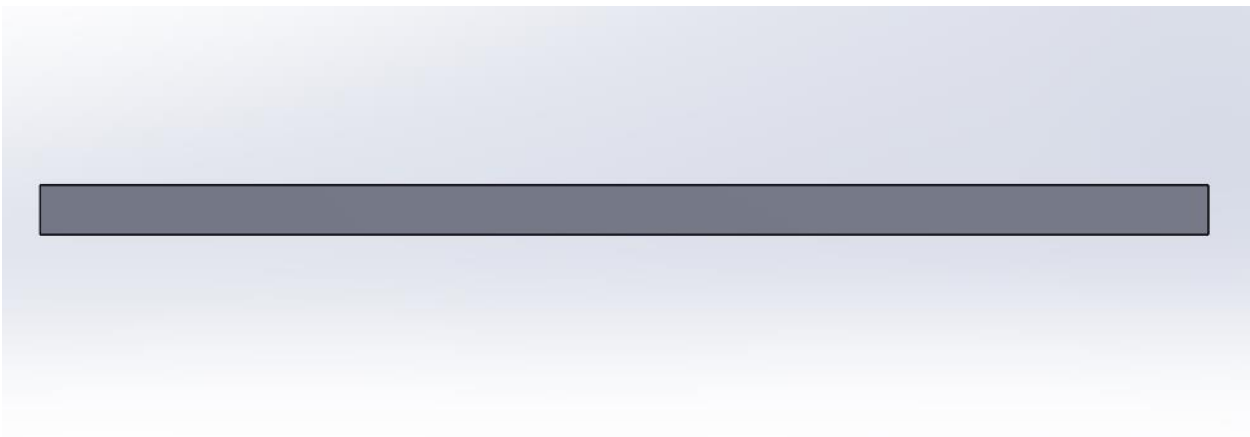
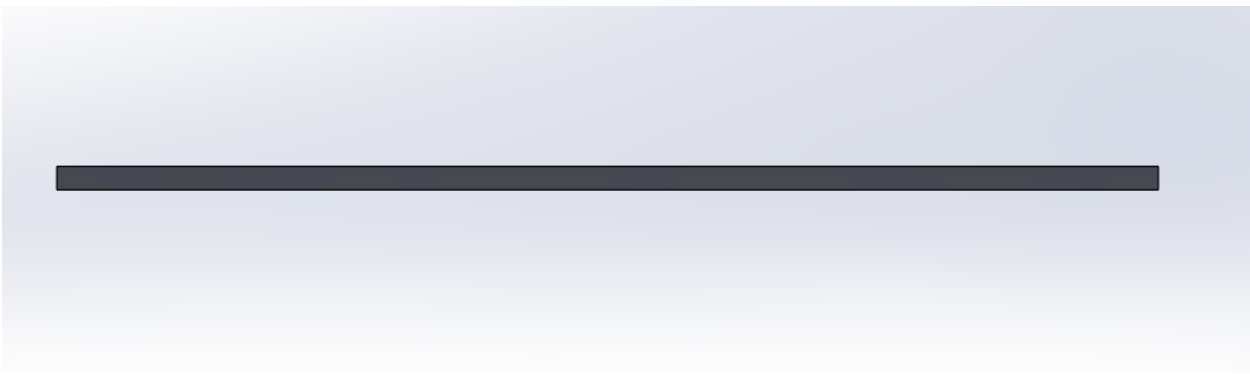
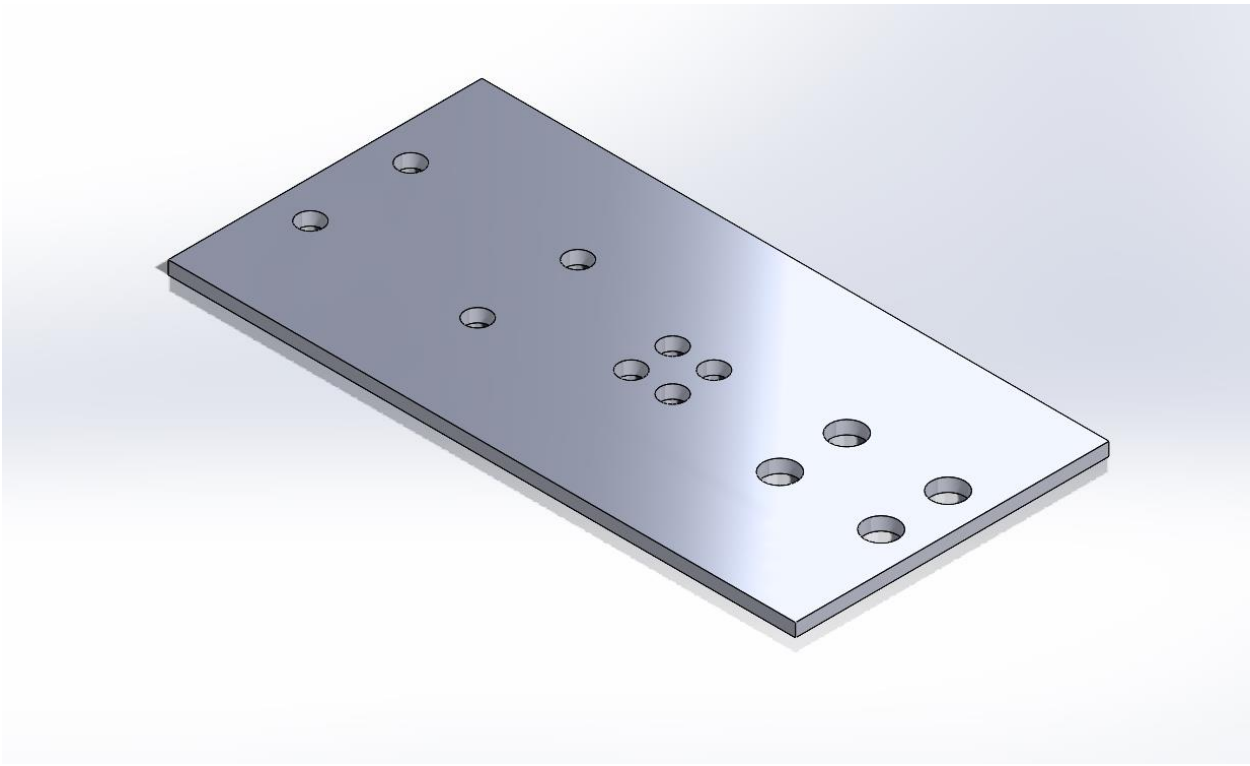
(units in m)

CAD:



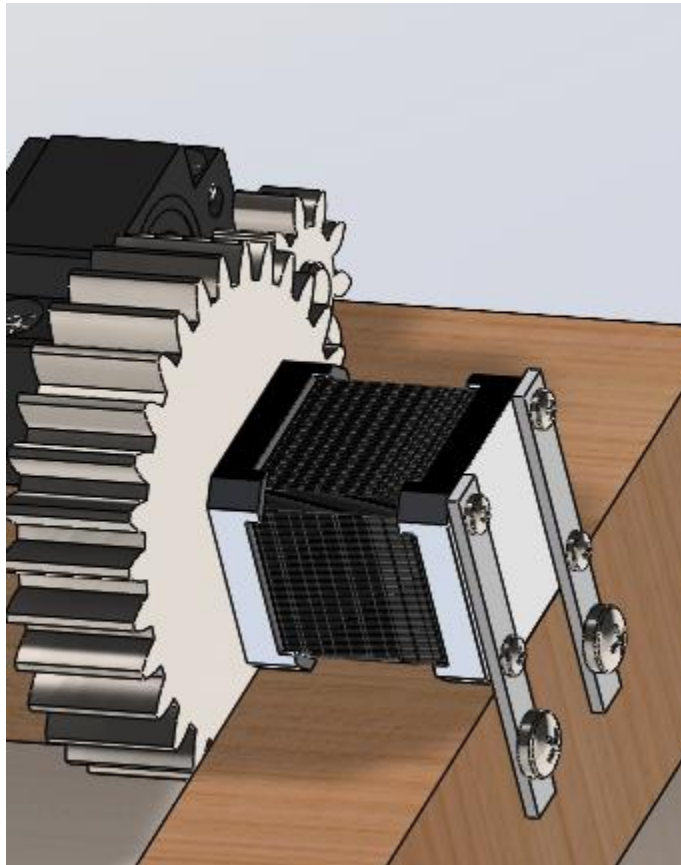
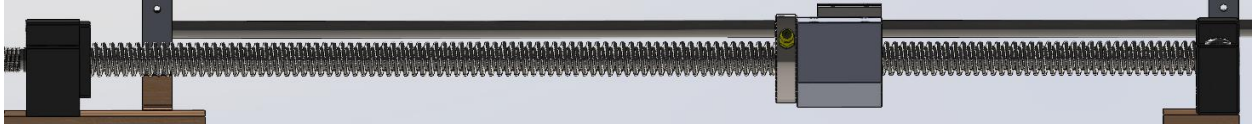
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: 1 LINEAR: 1 ANGULAR: 3			FINISH:		DEBUR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION
NAME	SIGNATURE	DATE							
DESIGN	Byron Borchardt	03/21/24							
CHECKED	Byron Borchardt	03/21/24							
APPROVED	Byron Borchardt	03/21/24							
DATE									
Q.A.									
			MATERIAL:				DWG. NO.		
			Aluminum 6061-O				base-01		
			WEIGHT: 97.22 g				SCALE: 1:1		SHEET 1 OF 1

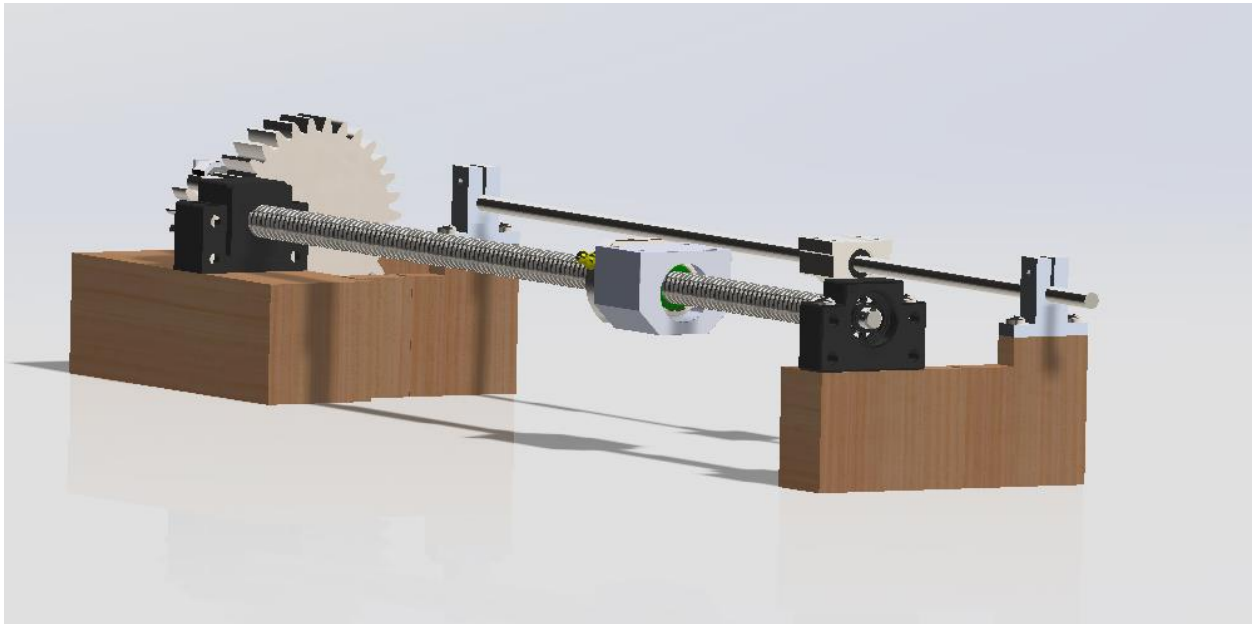




Linear Axis

Configuration

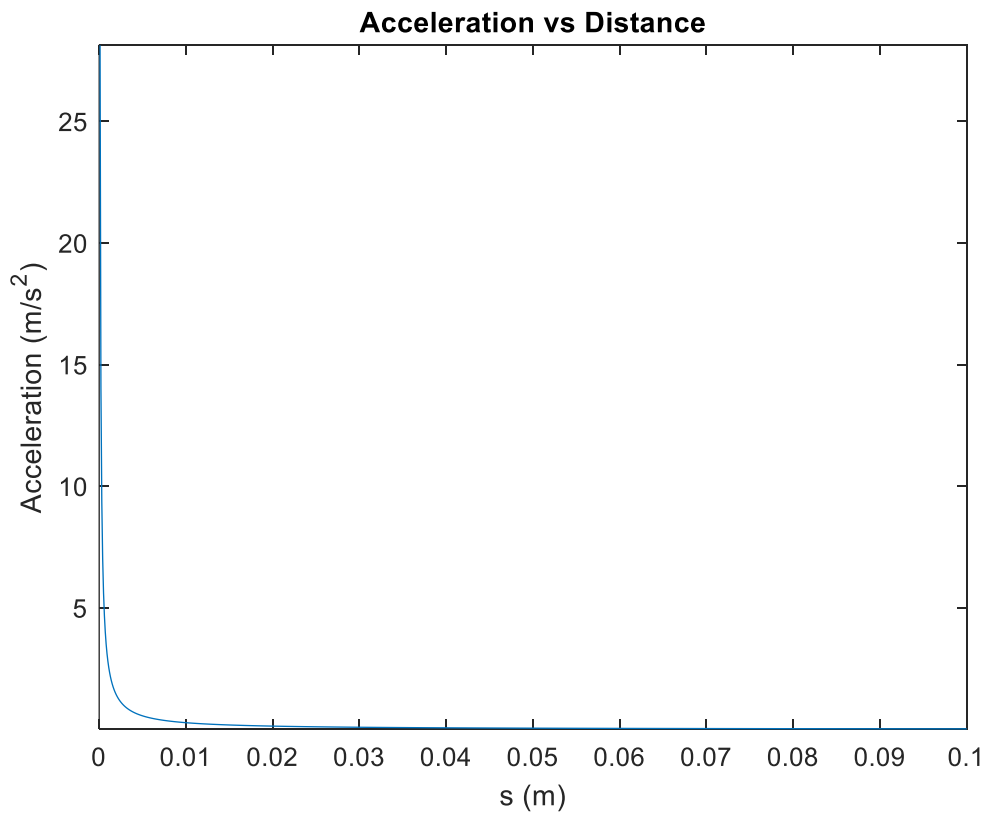


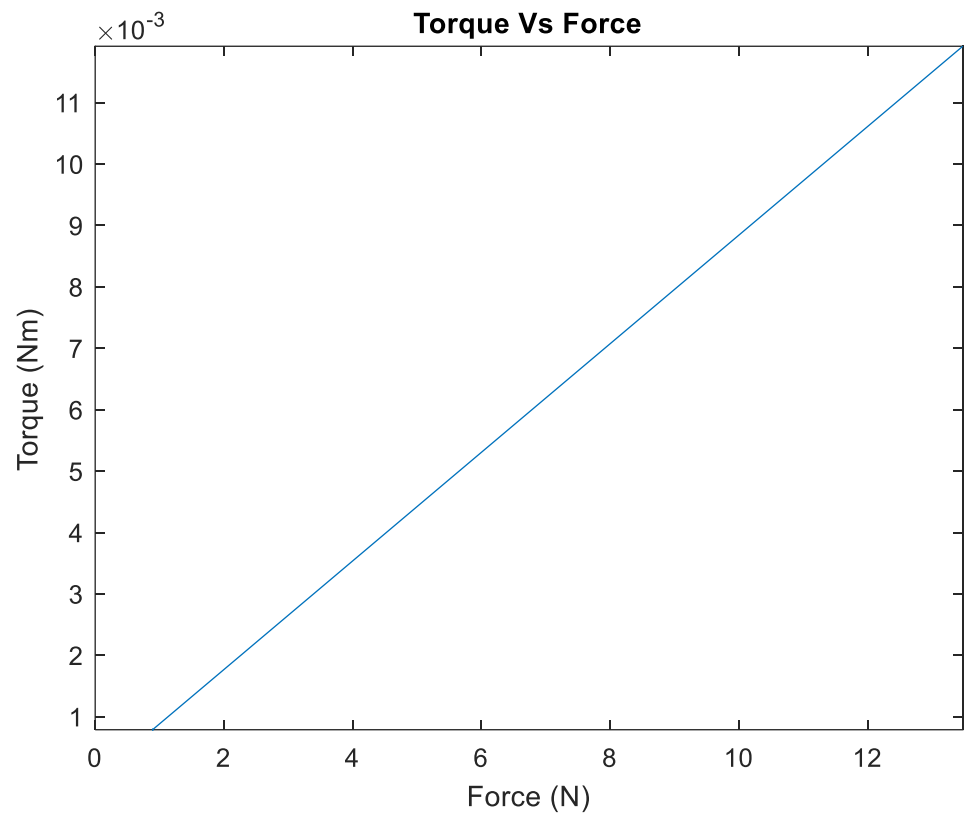


Equations

Torque

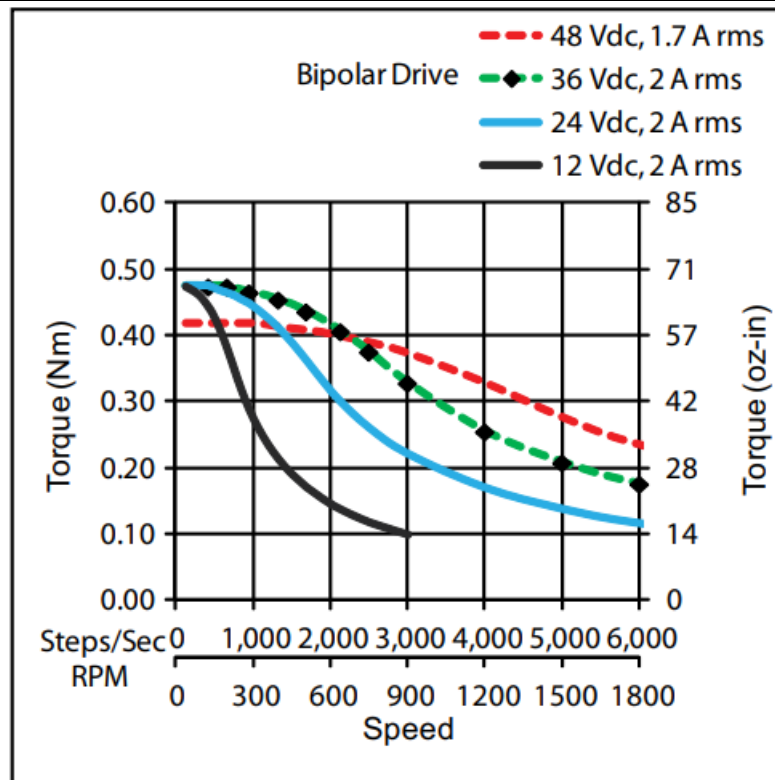
Variable	Equation	Required Spec
T	$\frac{F * L}{2\pi e} (GR)$	.036Nm
F	$m(a + 2\mu g)$	13.5N
a	$\frac{v^2}{2s}$	28 m/s ²
L	Lead Screw Spec	5mm/rev
μ	bearing friction	0.1
m	All components above linear axis	.448kg





NEMA 17 Motor Output

Torque@900rpm	.10Nm (holding) .045Nm (Position)
---------------	-----------------------------------

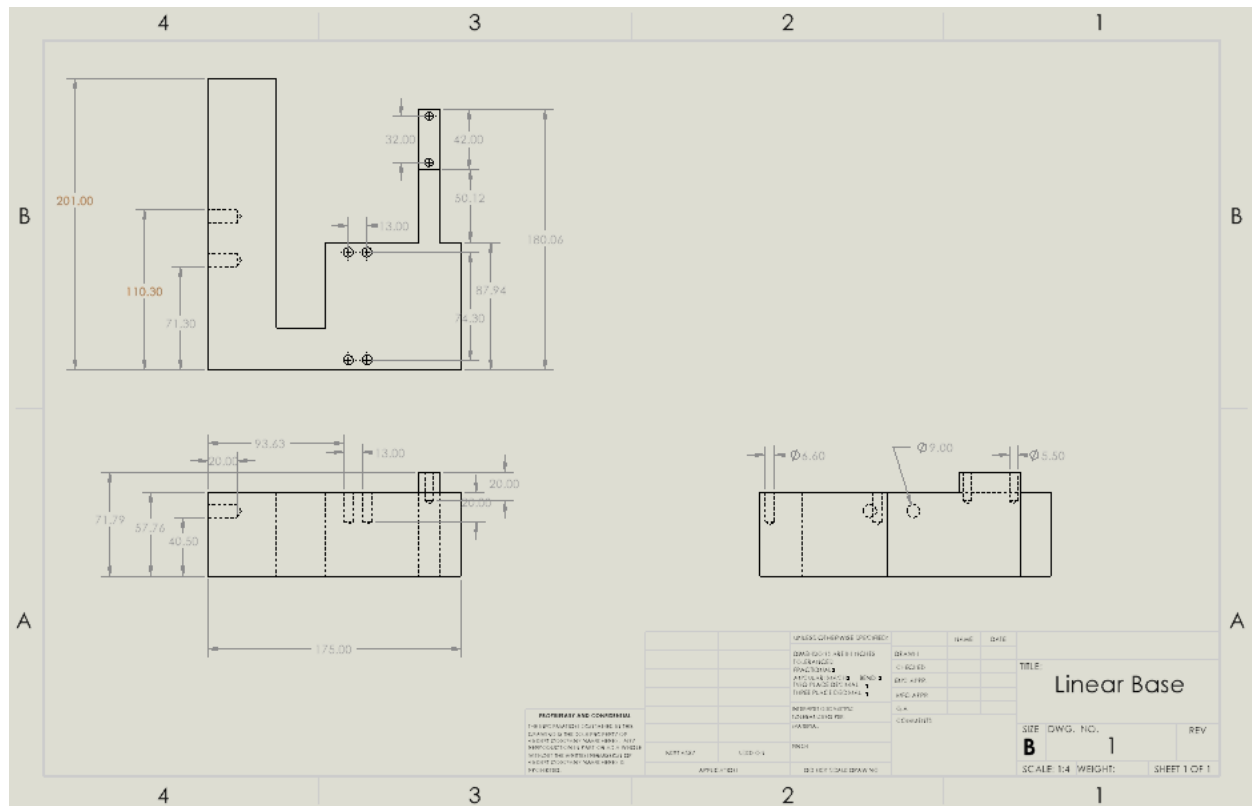


Credit: PBCLinear [Data-Sheet-Stepper-Motor-Support.pdf](http://pbclinear.com/Data-Sheet-Stepper-Motor-Support.pdf) (pbclinear.com)

Speed and resolution

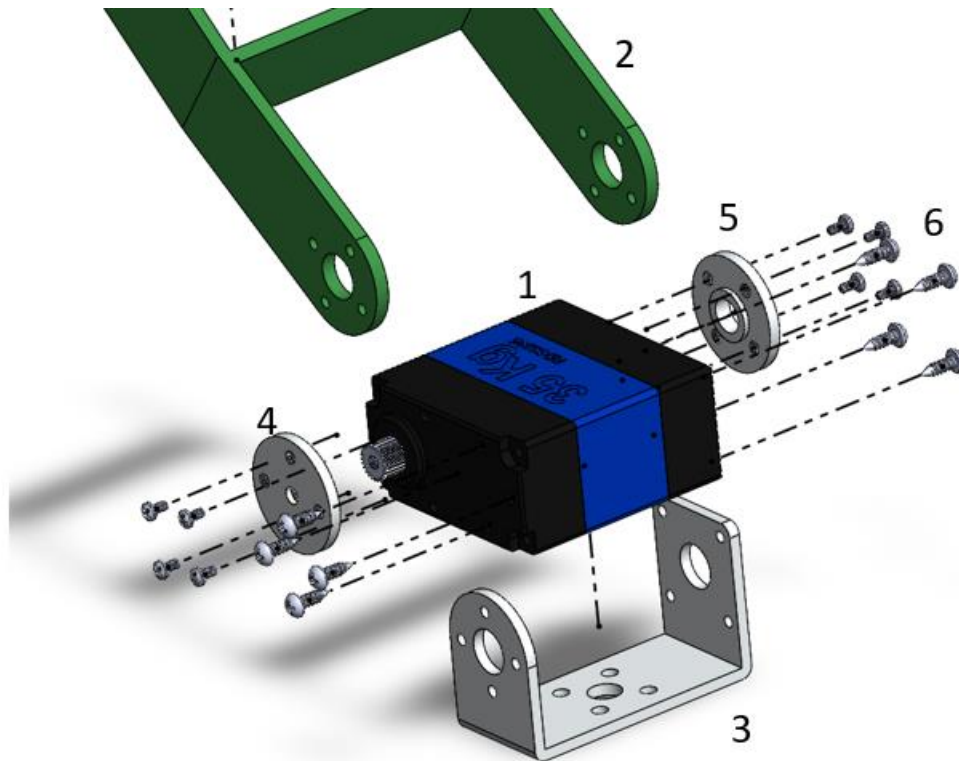
v	$(resolution * \frac{steps}{s})(GR)$	.225 m/s
Res.	$\frac{5mm}{200steps}$	.025mm
$\frac{steps}{s}$	Stepper motor @900rpm	3000 steps/s
GR	$\frac{Desired}{Max Motor rpm}$	3

Mass (of the arm on the linear axis)

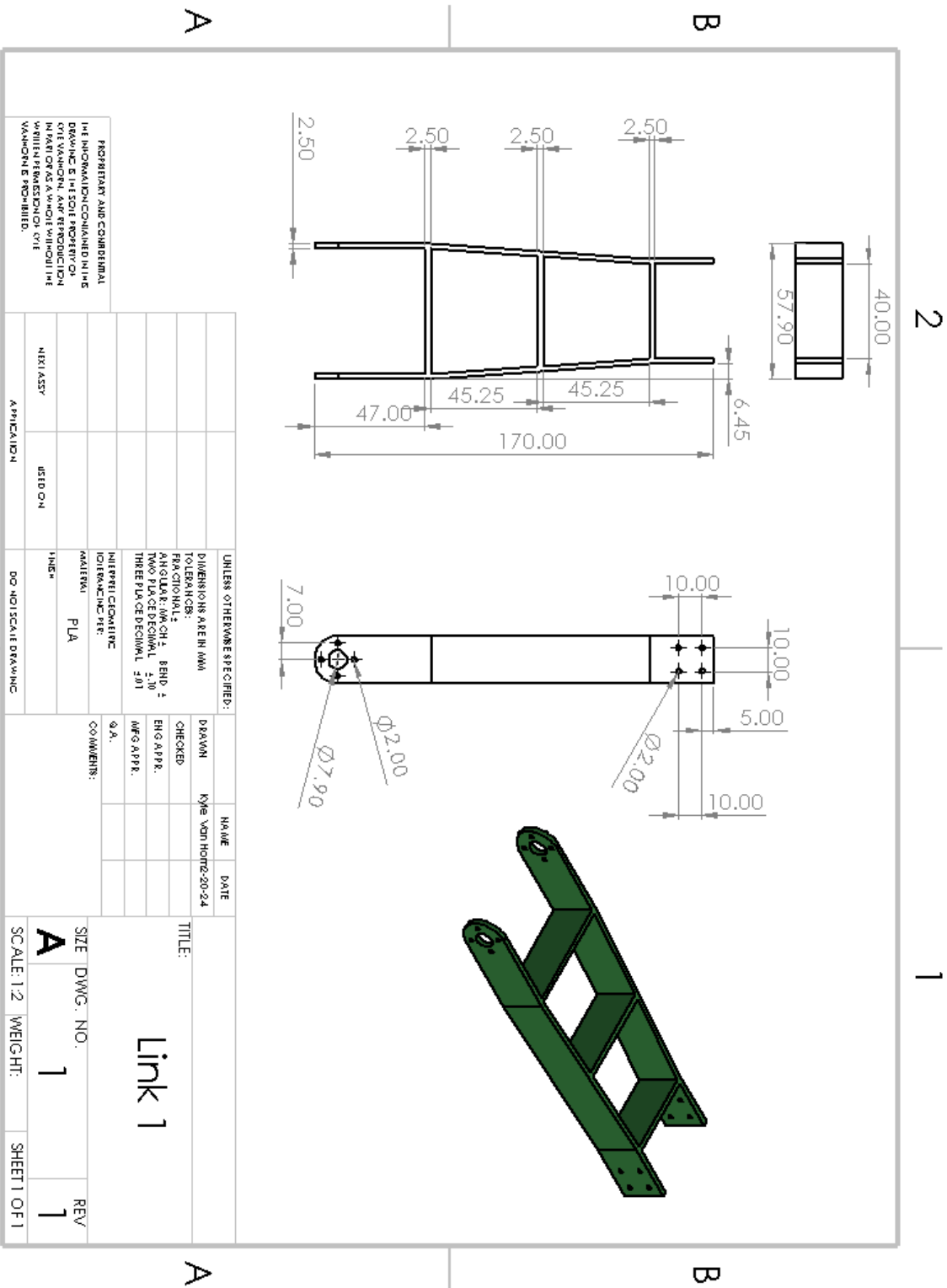


Rotary Axis 1



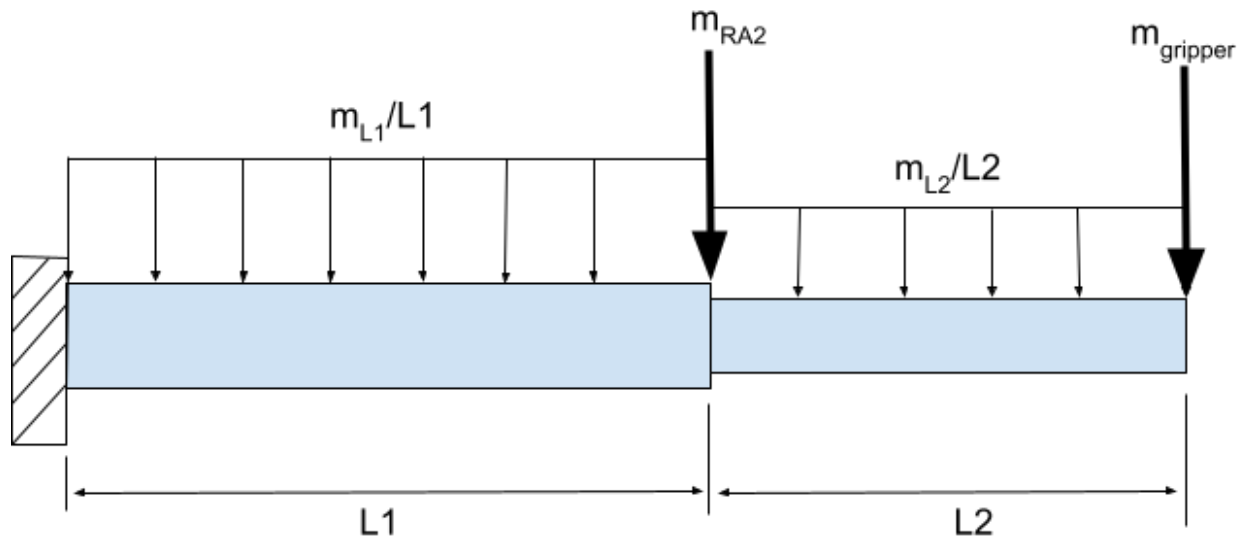


No.	Component	Qty
1	Double Bearing Servo Motor (35 kg)	1
2	3D Printed Custom Link 1	1
3	Aluminum U-Bracket Mount	1
4	Aluminum Gear Mounted Arm Plate	1
5	Aluminum Arm Plate	1
6	Screws (Included with Servo)	13



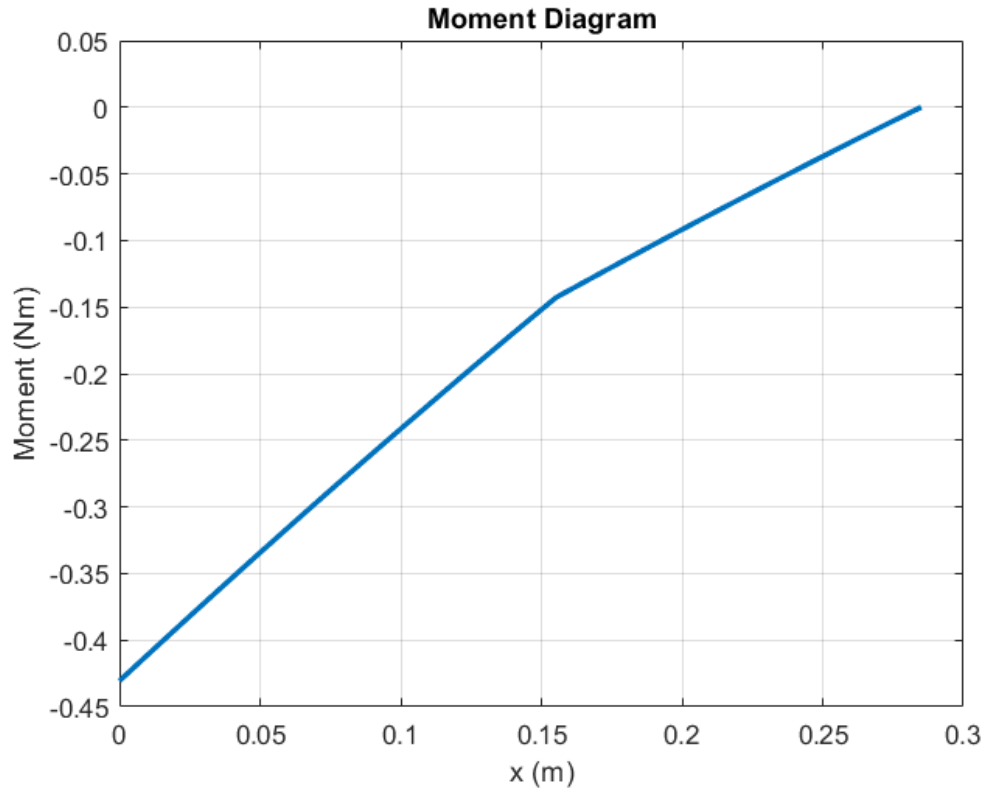
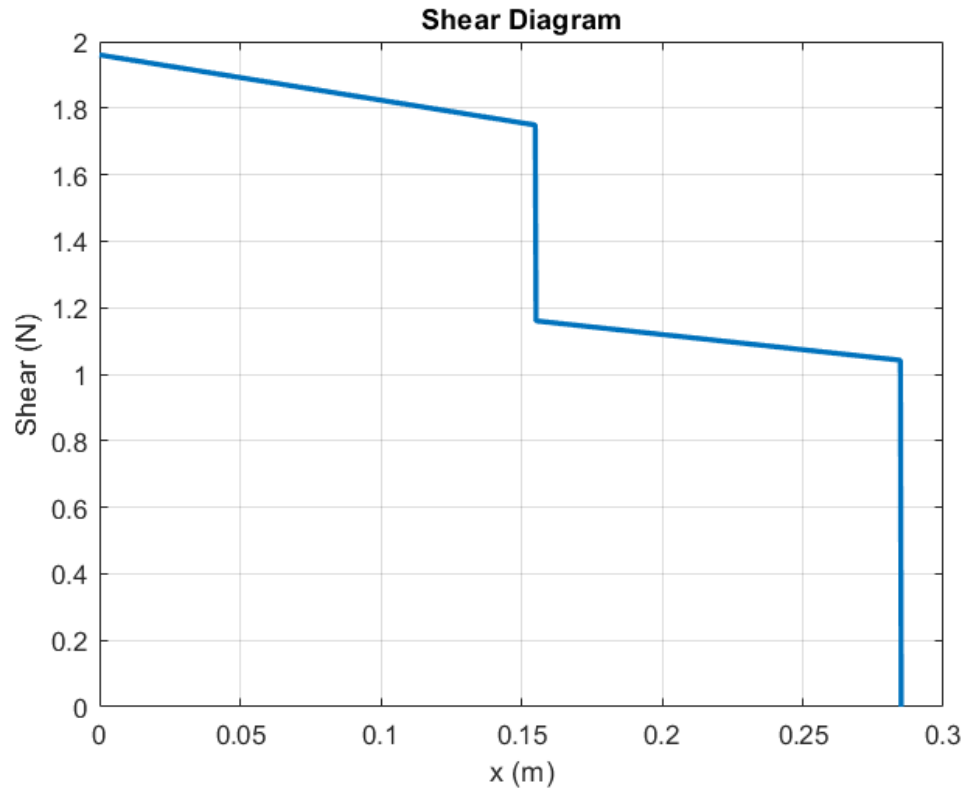
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DIMENSIONS ARE IN MM	DRAWN	NAME	DATE
TOLERANCES:	CHECKED	Kyle von Hontz-20-24	
FRACTIONAL:	ENG APPR.		
ANGULAR: MM CH 1 BEND 2	MFG APPR.		
TWO PLACE DECIMAL 2.00			
THREE PLACE DECIMAL 3.00			
INTERPRETATION:	Q.A.		
CONSTRUCTION:	COMMENTS:		
MATERIAL:			
FINISH:			
PLA			
NEXT ASSTY	DESIGN		
APPLICATION			
DO NOT SCALE DRAWING			



Variable	Description	Value
$L1$	Length of Link 1	15.5 cm
$L2$	Length of Link 2	13 cm
m_{L1}	Mass of Link 1	21.5 g
m_{L2}	Mass of Link 2	12.1 g
m_{RA2}	Mass of RA2	60 g
$m_{gripper}$	Mass of Gripper	103.2 g

Section (m)	Shear (N)	Moment (Ncm)
$0 \leq x \leq 0.155$	$V = 1.959 - 1.36x$	$M = 1.959x - 0.68x^2 - 0.43$
$0.155 \leq x \leq 0.285$	$V = 1.302 - 0.913x$	$M = 1.302x - 0.456x^2 - 0.334$



Bending Stress

Cross Section = 5 mm [a] x 20 mm [b]

$$\sigma = \frac{Mc}{I}$$

Variable (Inputs)	Definition	Value
M	Max Bending Moment	43.05 Ncm
c	Distance to neutral axis (b/2)	10 mm
I	Area Moment of Inertia (ab ³ /12)	3333.3 mm ⁴
Variable (Outputs)	Definition	Value
σ	Bending Stress	1.29 MPa

Factor of Safety

$$FS = \frac{\sigma_y}{\sigma}$$

Variable (Inputs)	Definition	Value
σ	Bending Stress	1.29 MPa
σ _y	Yield Strength of PLA	26 MPa
Variable (Outputs)	Definition	Value
FS	Factor of Safety	20

Shear Stress

$$\tau_{max} = \frac{3V}{2A}$$

Variable (Inputs)	Definition	Value
V	Max Shear Force	1.959 N
A	Cross Sectional Area	100 mm
Variable (Outputs)	Definition	Value
τ _{max}	Max Shear Stress	0.029 MPa

Torque Requirements for RA1

$$J_{total} = J_{L1} + J_{L2} + J_{RA2} + J_{gripper}$$

Where:

$$J_{L1} = \frac{m_{L1} * b^2}{12} + \frac{m_{L1} * L_1^2}{3}$$

$$J_{L2} = \frac{1}{12} m_{L2} * (w^2 + d^2) + m_{L2} \left(L_1 + \frac{L_2}{2} \right)^2$$

$$J_{RA2} = m_{RA2} * L_1^2$$

$$J_{gripper} = m_{gripper} * (L_1 + L_2)^2$$

Variable (Inputs)	Definition	Value
m_{L1}	Mass of link 1	0.0215 kg
b	Height of link 1	20 mm
L_1	Length of link 1	15.5 cm
m_{L2}	Mass of link 2	0.0121 kg
w	Width of link 2	4 mm
d	Depth of link 2	20 mm
L_2	Length of link 2	13 cm
m_{RA2}	Mass of RA2	60 g
$m_{gripper}$	Mass of gripper	106.2 g
Variable (Outputs)	Definition	Value
J_{L1}	Inertia of Link 1 about RA1	$1.729 * 10^{-4} \text{ kg} * \text{m}^2$
J_{L2}	Inertia of Link 2 about RA1	$5.861 * 10^{-4} \text{ kg} * \text{m}^2$
J_{RA2}	Inertia of RA2 about RA1	$0.001 \text{ kg} * \text{m}^2$
$J_{gripper}$	Inertia of Gripper about RA1	$0.009 \text{ kg} * \text{m}^2$
J_{total}	Total Mass Moment of Inertia about RA1	$0.011 \text{ kg} * \text{m}^2$

Gravitational Load

$$L = \left(m_{L1} * \frac{L_1}{2} + m_{L2} * \left(L_1 + \frac{L_2}{2} \right) + m_{RA2} * L_1 + m_{gripper} * (L_1 + L_2) \right) * g$$

Where the variables are the same as before and g is gravity

Variable (Outputs)	Definition	Value
L	Gravitational Load	43.047 Ncm

Total Required Torque for RA1

$$T = J_{total} * \alpha + L$$

Variable (Inputs)	Definition	Value
J_{total}	Total Mass Moment of Inertia about RA1	$0.011 \text{ kg} * \text{m}^2$
α	Angular acceleration	52.4 rad/s^2
L	Gravitational Load	43.047 Ncm
Variable (Output)	Definition	Value
T	Total Required Torque	99.778 Ncm

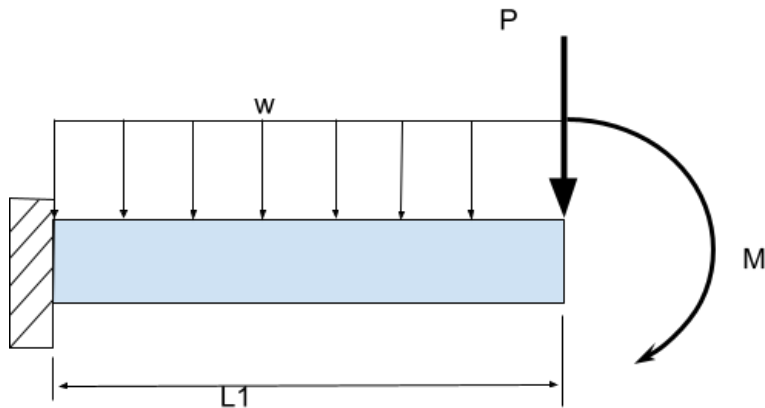
Resolution Calculations

Achievable Resolution with Servo Motor: 0.5°

Total Link Length Actuated by RA1 = 28.5 cm

End Effector Resolution = $28.5 \sin(0.5^\circ) = 0.25 \text{ cm}$

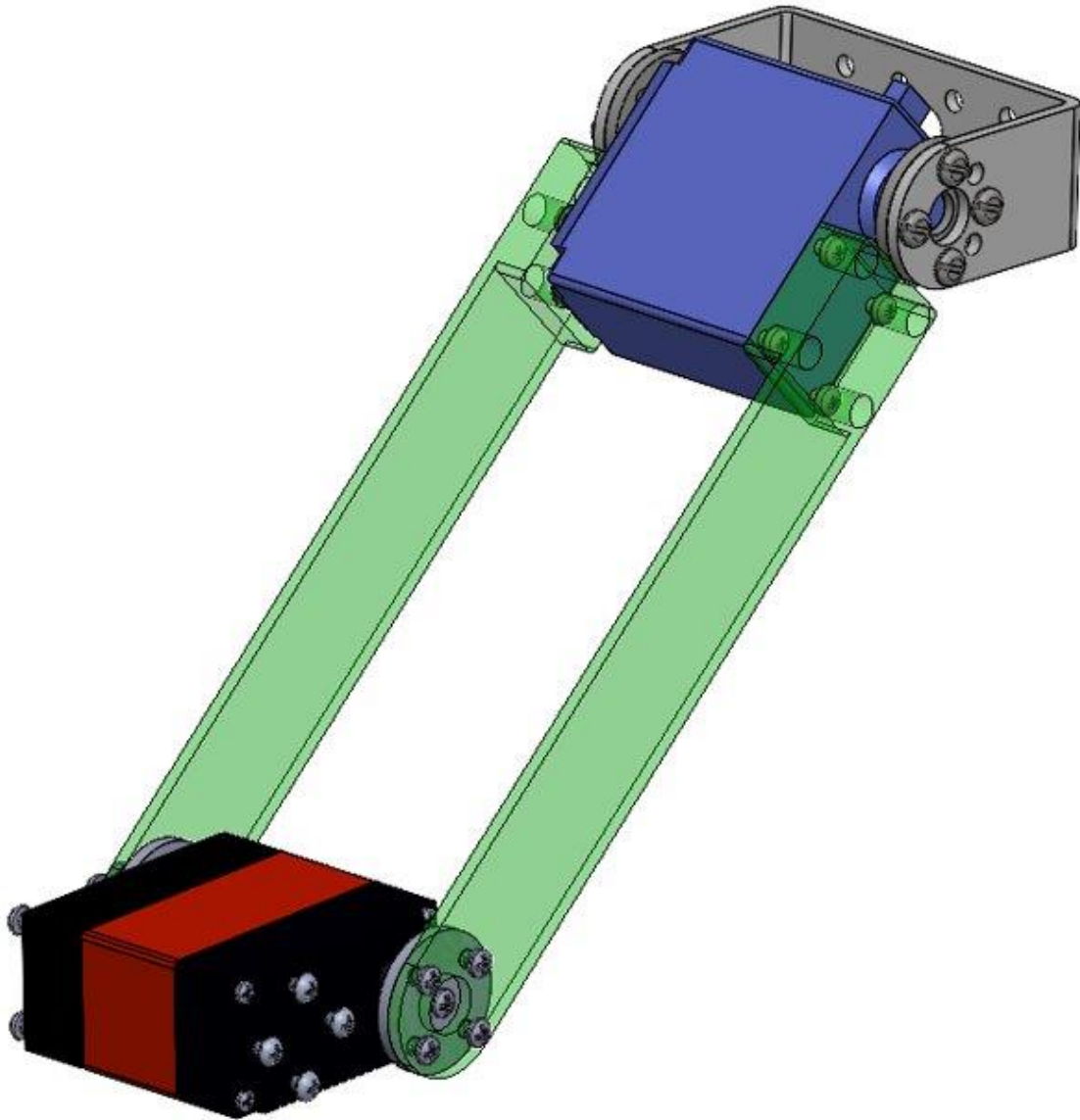
Deflection at the End of Link 1

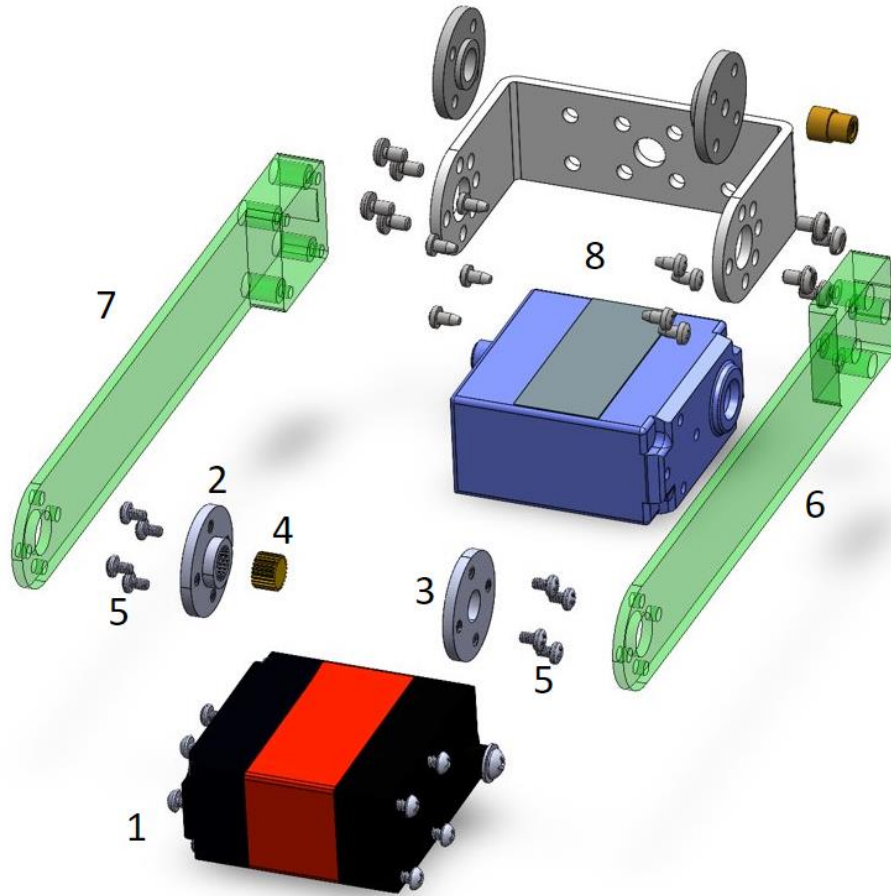


$$v_{max} = \frac{-PL^3}{3EI} - \frac{ML^2}{2EI} - \frac{wL^4}{8EI}$$

Variable (Input)	Description	Value
w	Distributed Load of Link 1	1.36 N/m
P	Load of RA2, Link 2, and Gripper	1.749 N
M	Moment Caused by RA2, Link 2, and Gripper about RA1	0.414 Nm
L1	Length of Link 1	15.5 cm
E	Young's Modulus of PLA	3.5 GPa
I	Area Moment of Inertia	3333 mm ⁴
Variable (Output)	Description	Value
v	Deflection at the end of link 1	0.621 mm

Rotary Axis 2

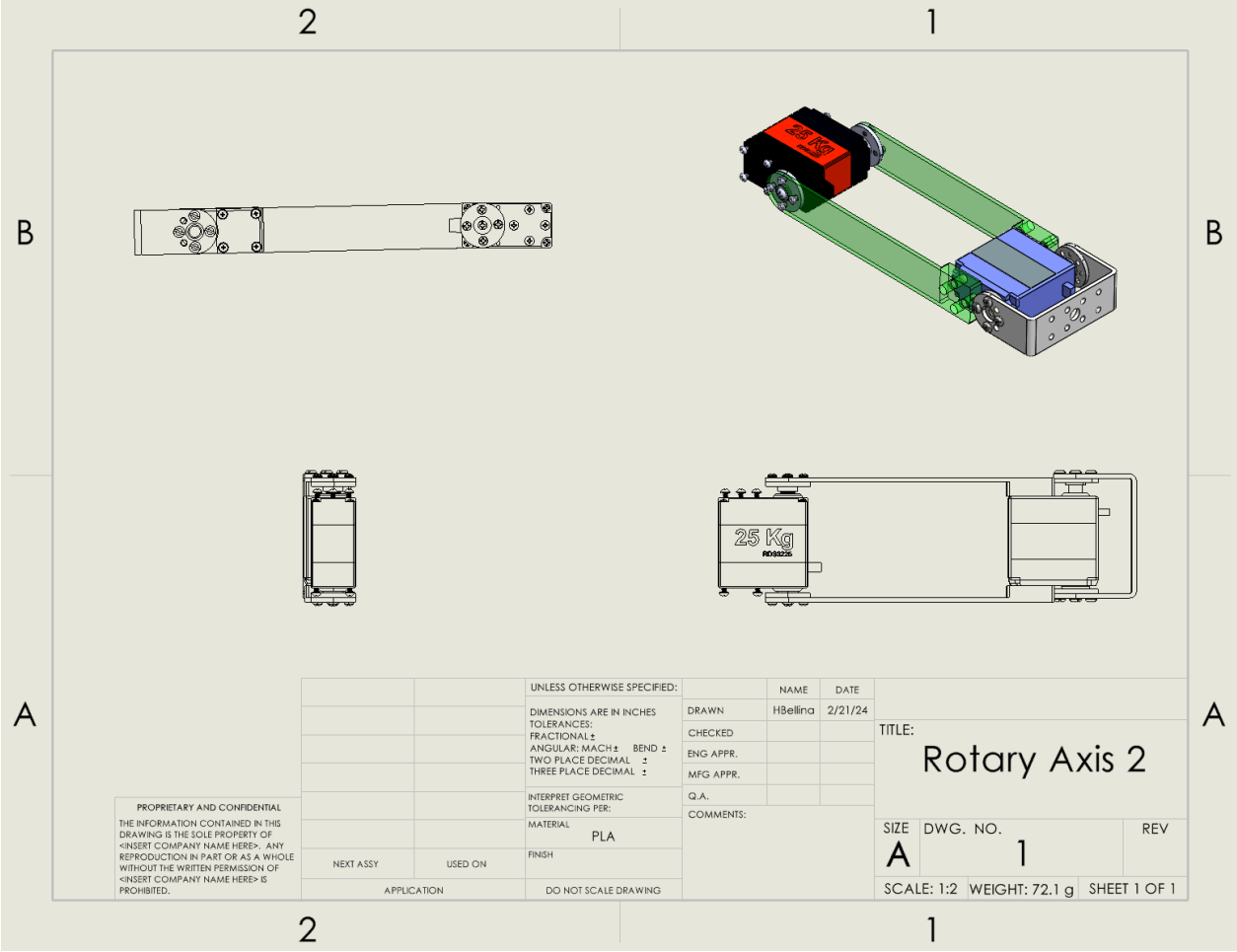


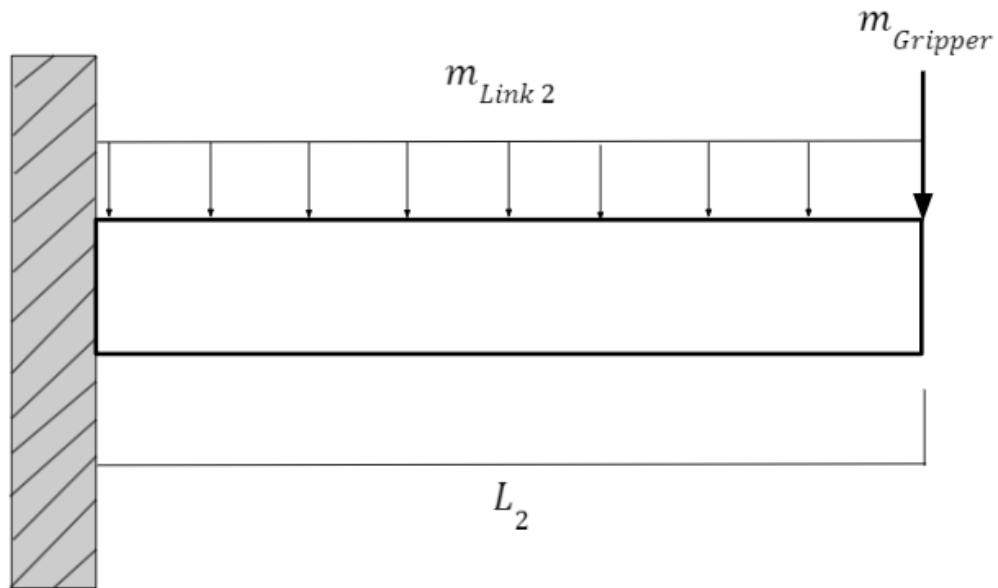


No.	Component	Qty
1	Double Bearing Servo Motor (20 kg)	1
2	Gear Side Aluminum Mounting Disk*	1
3	Aluminum Mounting Disk*	1
4	Servo Motor Direct Drive Gear*	1
5	Mounting Screws for Link Arms*	8
6	Right Side Link Arm	1
7	Left Side Link Arm	1
8	Gripper Actuation Servo **	1

*These components come with RA 2 Servo.

**This is not included in RA 2 from an analytical standpoint, and rather is contained within the point load for the gripper. In this instance this servo is used as a visual aid in how it will couple with RA2/Link 2. During modeling, the Gripper Actuation Servo was also constrained into RA2 to make assembly movement easier, so that is why it can be seen here.

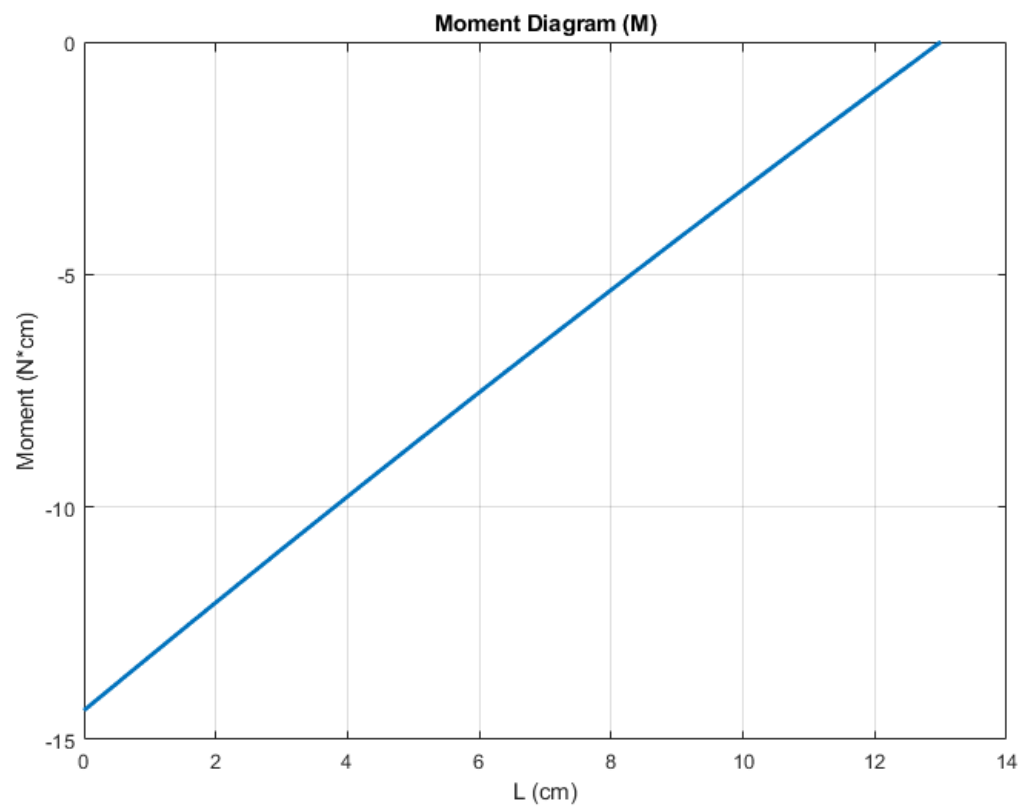
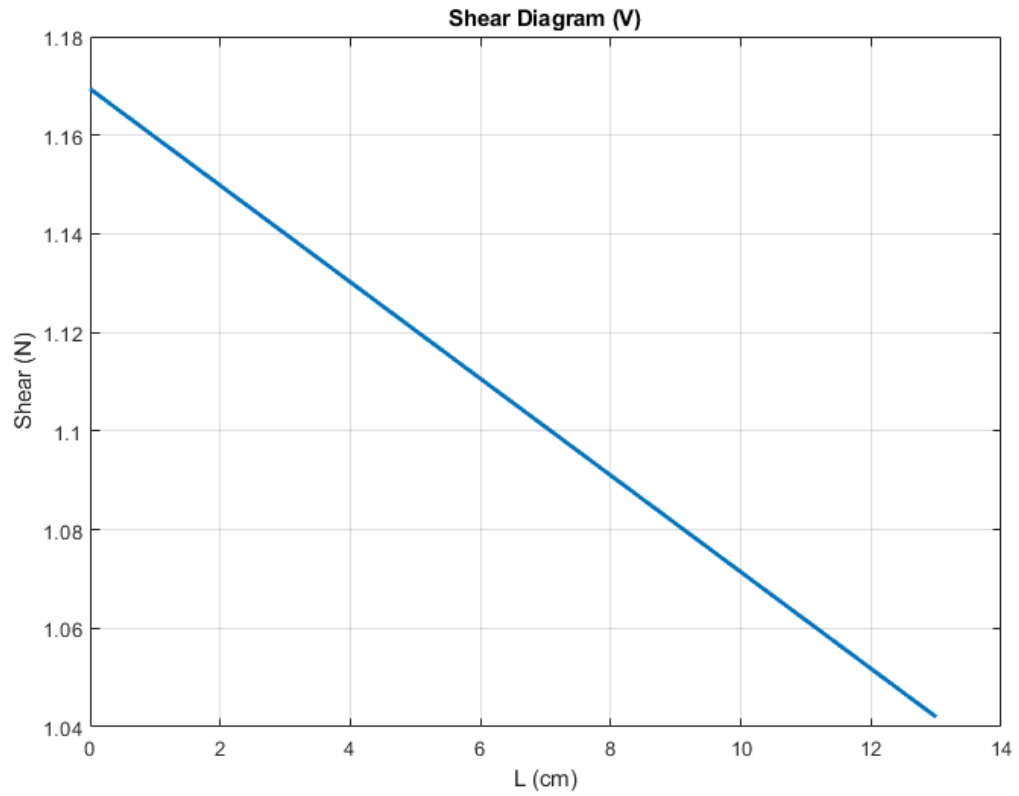




Variable	Description	Value
L2	Length of Link 2	13 cm
$m_{Link\ 2}$	Mass of Link 2	12.1 g
$m_{gripper}$	Mass of Gripper	106.2 g

Section (m)	Shear (N)	Moment (Ncm)
$0 \leq x \leq 13$	$1.1694 - 0.0098x$	$1.1694x - 0.0049x^2 - 14.3741$

Shear and Moment Diagrams



Bending Stress

Cross Section = 4 mm [a] x 20 mm [b]

$$\sigma = \frac{Mc}{I}$$

Variable (Inputs)	Definition	Value
M	Max Bending Moment	14.3741 Nm
c	Distance to neutral axis (b/2)	10 mm
I	Area Moment of Inertia (ab ³ /12)	2666.7mm ⁴
Variable (Outputs)	Definition	Value
σ	Bending Stress	0.5390 MPa

Factor of Safety

$$FS = \frac{\sigma_{YS}}{\sigma}$$

Variable (Inputs)	Definition	Value
σ	Bending Stress	0.54 MPa
σ _{YS}	Yield Strength of PLA	26 MPa
Variable (Outputs)	Definition	Value
FS	Factor of Safety	48.2

Shear Stress

$$\tau = \frac{3V}{2A}$$

Variable (Inputs)	Definition	Value
V	Max Shear Force	1.1694 N
A	Cross Sectional Area	40 mm ²
Variable (Outputs)	Definition	Value
τ	Max Shear Stress	0.0219 MPa

Torque Requirements for RA2

$$J_z(Total) = J_z(PointMass) + J_z(Link2)$$

Where:

$$J_z(Point\ Mass) = m_g + L_2^2$$

$$J_z(Link\ 2) = \frac{m_2 b_2^2}{12} + \frac{m_2 L_2^2}{3}$$

Variable (Inputs)	Definition	Value
m_{L2}	Mass of link 2	12.1 g
m_g	Mass of gripper	106.2 g
b	Height of link 2 at cross section	20 mm
L_2	Length of link 2	130 mm
Variable (Outputs)	Definition	Value
$J_z(Link\ 2)$	Inertia of Link 2 about RA2	0.0018 mm^2
$J_z(Point\ Mass)$	Inertia of Gripper about RA2	$7.33 \cdot 10^{-7} mm^2$
$J_z(Total)$	Total Mass Moment of Inertia about RA2	0.0018 mm^2

Gravitational Load

$$L = \left(m_2 \frac{L_2}{2} + m_g L_2 \right) g$$

Where the variables are the same as before and g is gravity

Variable (Outputs)	Definition	Value
L	Gravitational Load	14.3726 Ncm

Total Required Torque for RA1

$$T = J_z(Total) \cdot \alpha + L$$

Variable (Inputs)	Definition	Value
$J_z(Total)$	Total Mass Moment of Inertia about RA1	0.0019 kgm^2
α	Angular acceleration	52.4 $\frac{rads}{s^2}$
L	Gravitational Load	0.1437 kgm
Variable (Output)	Definition	Value
T	Total Required Torque	24.16 Ncm

Resolution Calculations

Achievable Resolution with Servo Motor: 0.5°

Total Link Length Actuated by RA1 = 13 cm

End Effector Resolution = $13\sin(0.5^\circ) = 0.113$ cm

Deflection at the End of Link 2

$$v = -\frac{PL^3}{3EI} - \frac{ML^2}{2EI} - \frac{wL^2}{8EI}$$

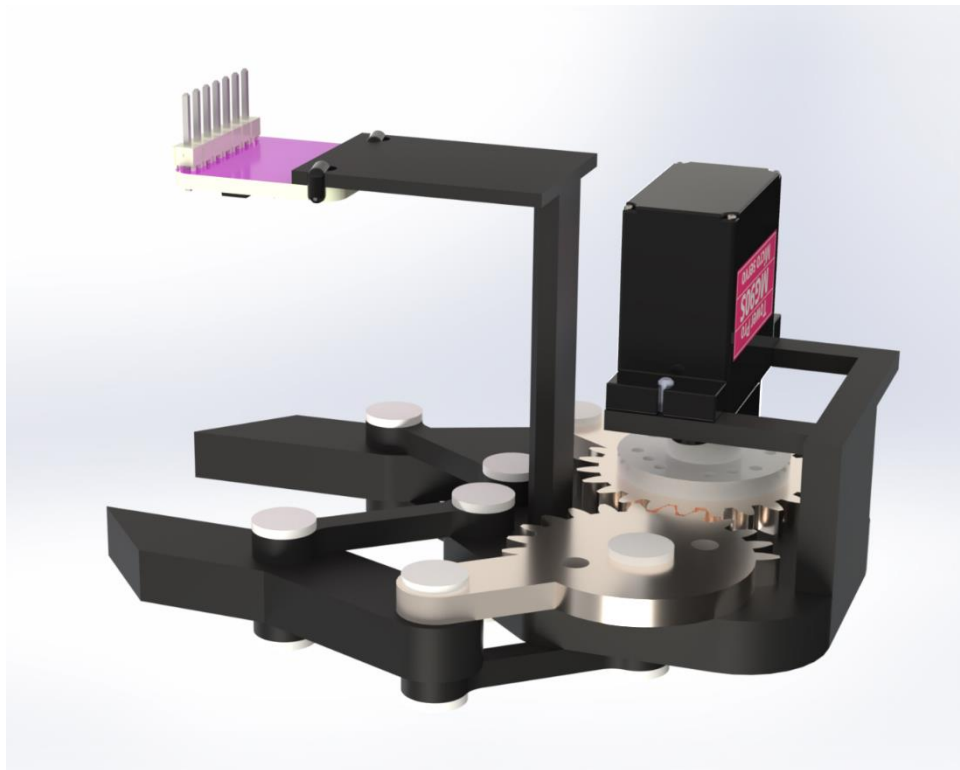
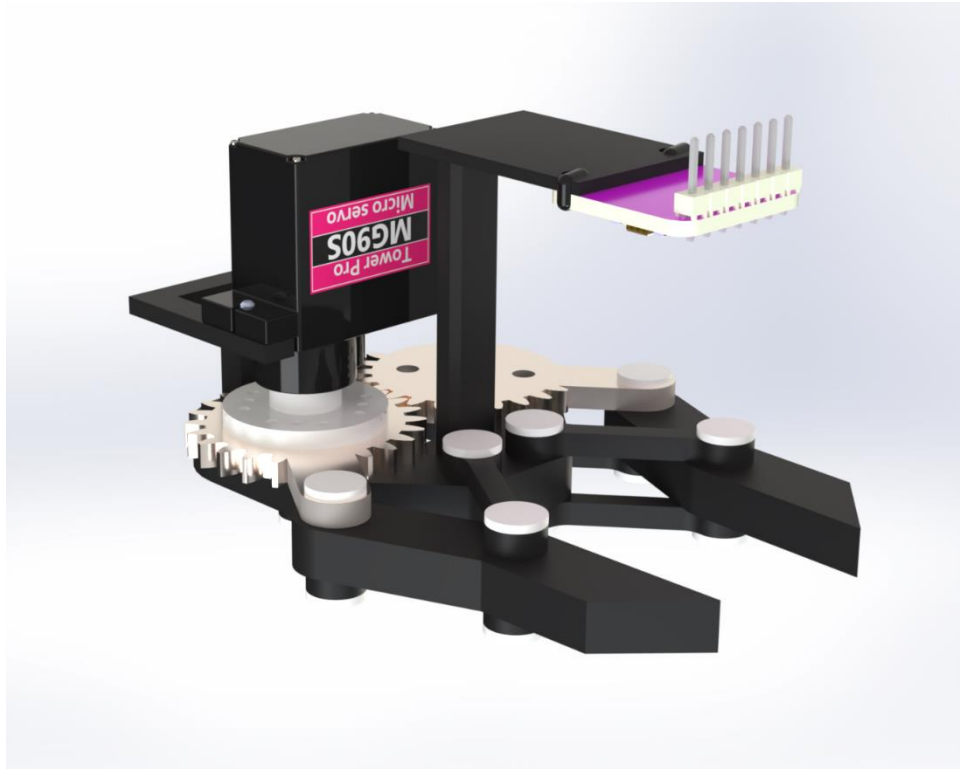
Variable (Input)	Description	Value
w	Distributed Load of Link 2	0.98 N/m
P	Load of Gripper	1.0418 N
M	Moment Caused by Gripper about RA2	0.135 Nm
L_2	Length of Link 2	13 cm
E	Young's Modulus of PLA	3.5 GPa
I	Area Moment of Inertia	2666.7 mm ⁴
Variable (Output)	Description	Value
v	Deflection at the end of link 2	0.0821 mm

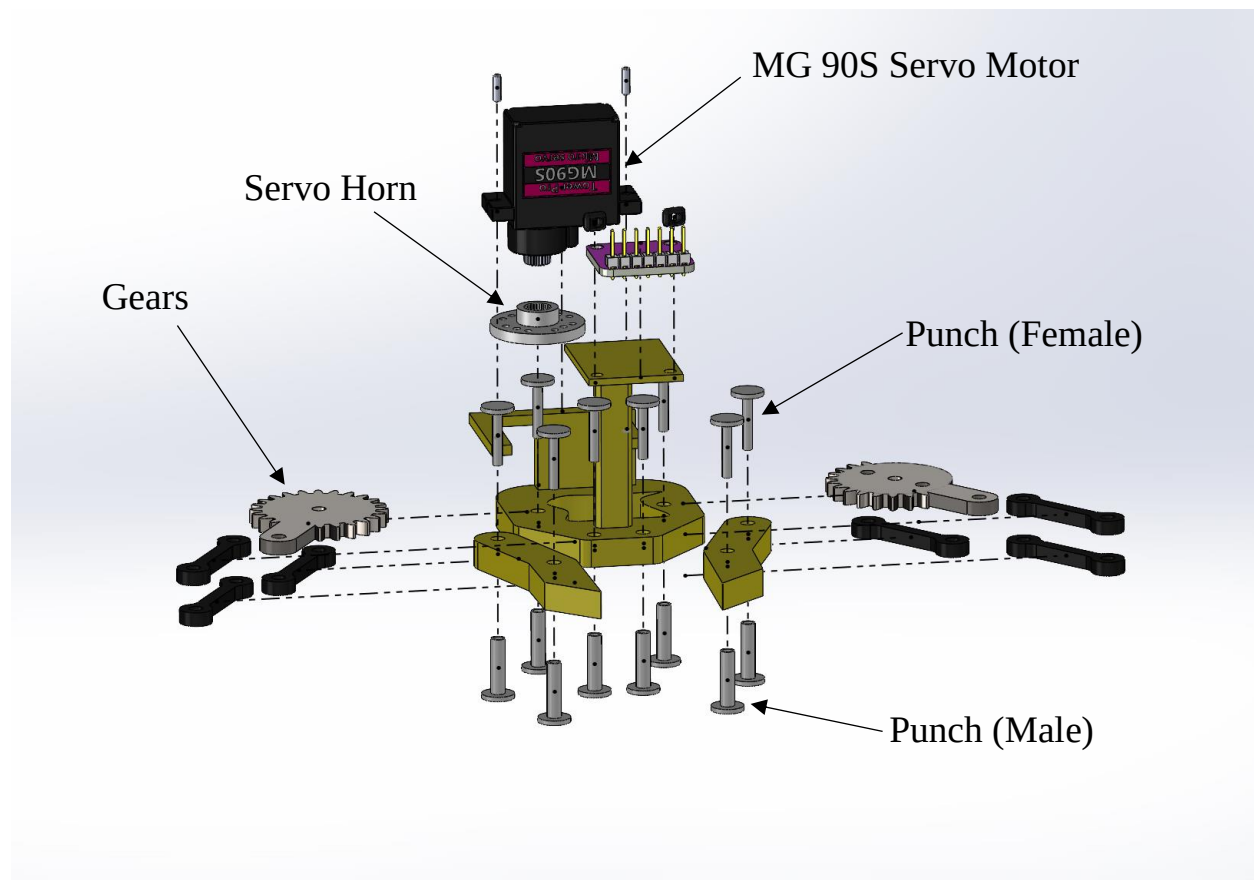
Therefore, it can be stated that the total deflection at the end of link two is as follows

$$v = v(\text{link 1}) + v(\text{link 2})$$

Variable (Input)	Description	Value
$v(\text{link 1})$	Deflection at the end of Link 1	0.621 mm
$v(\text{link 2})$	Deflection at the end of Link 2	0.0821 mm
Variable (Output)	Description	Value
v	Total deflection at the end of link 2 with respect to link one	0.7031 mm

Gripper





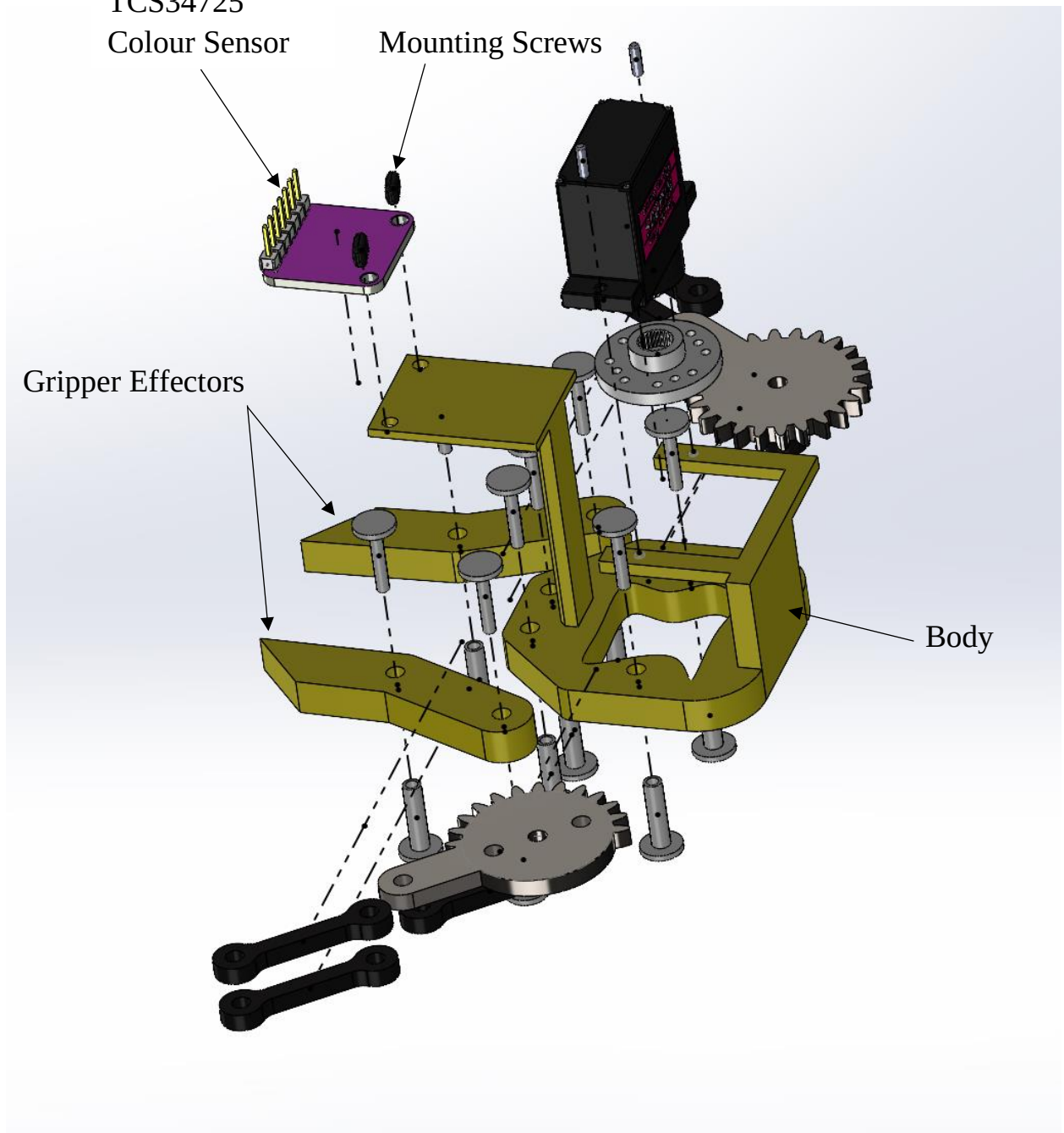
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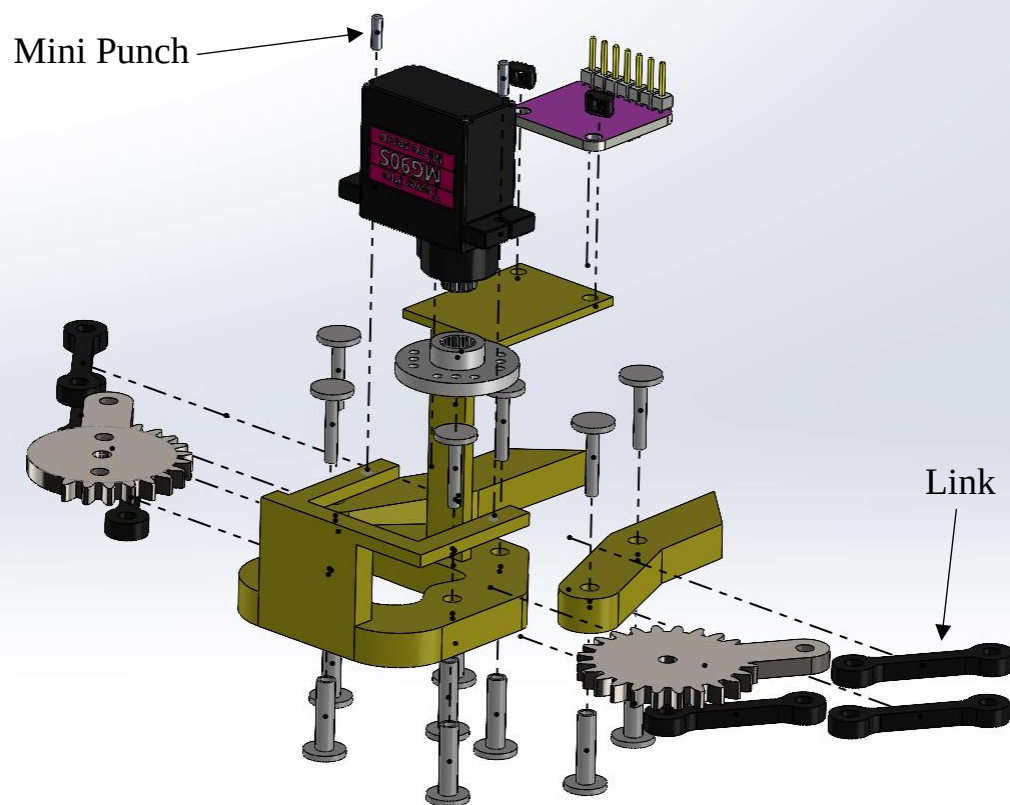
Colour Sensor

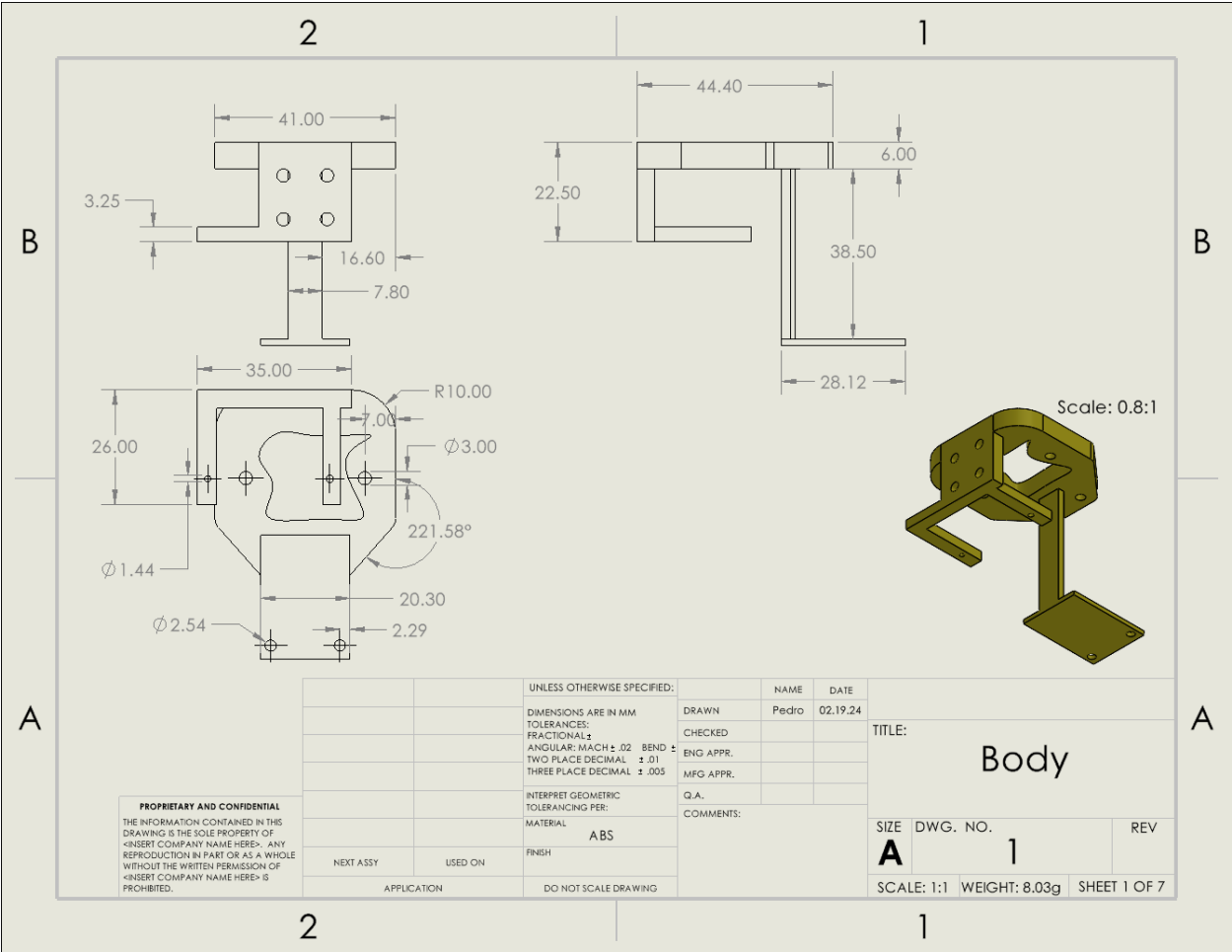
Mounting Screws

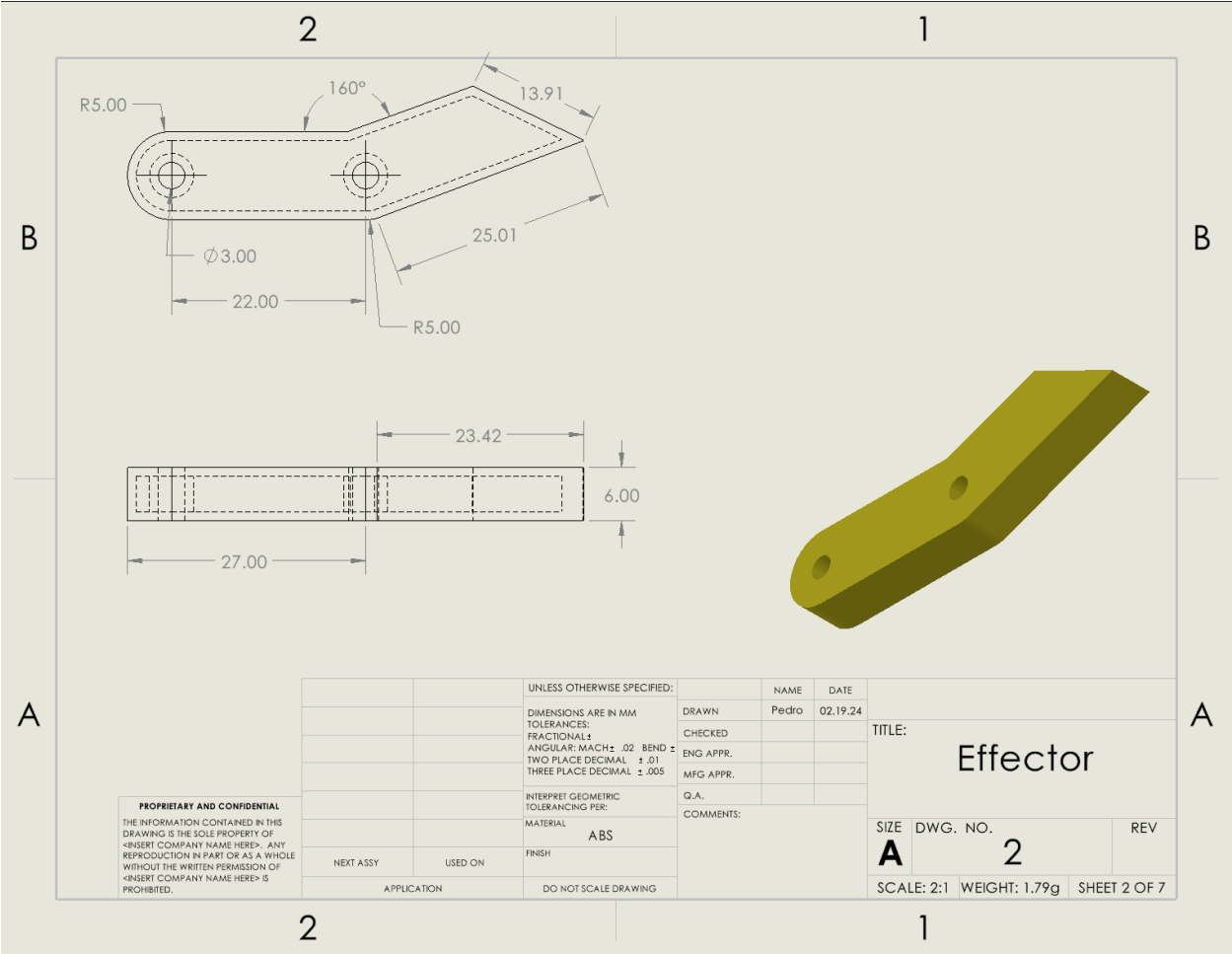
Gripper Effectors

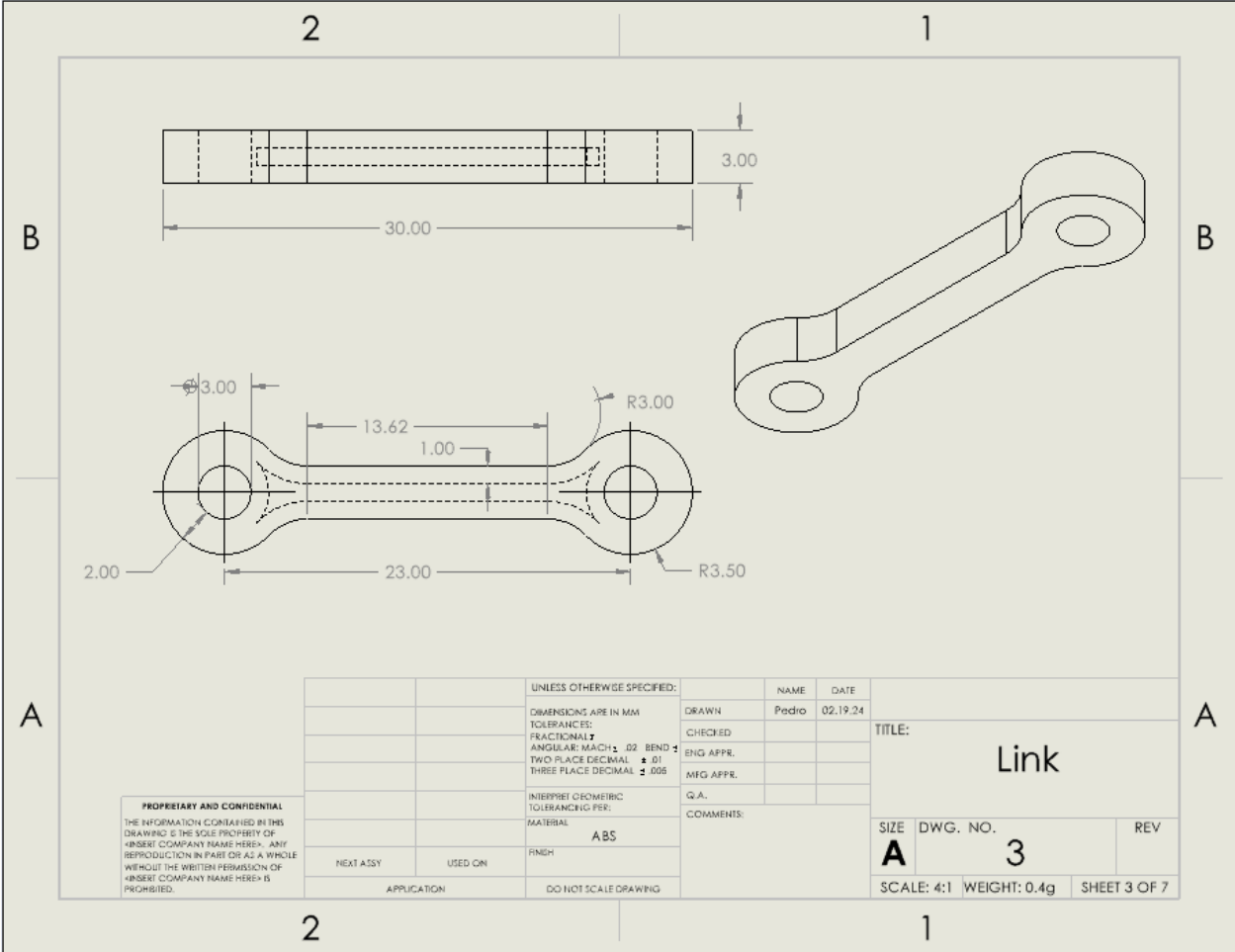
Body

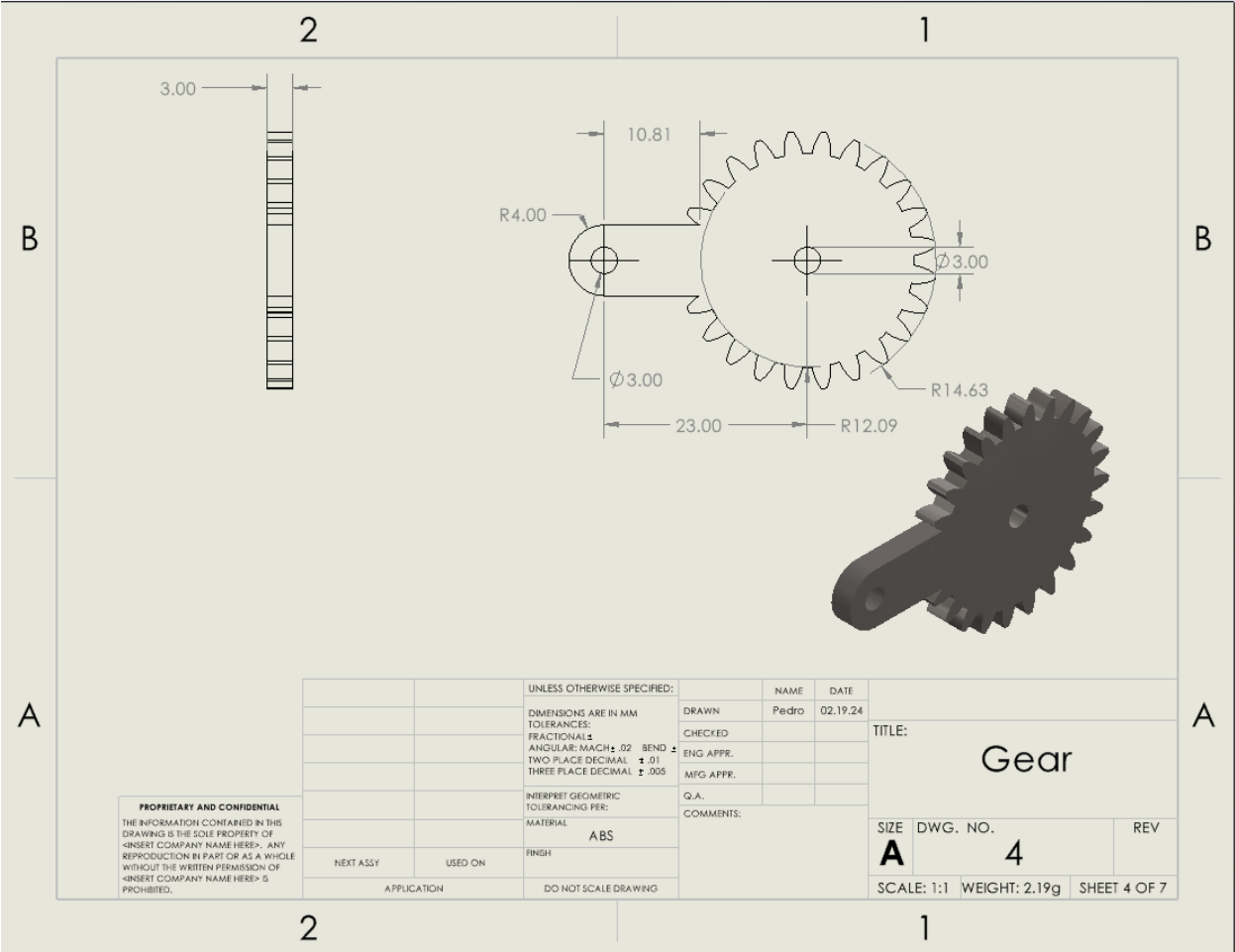


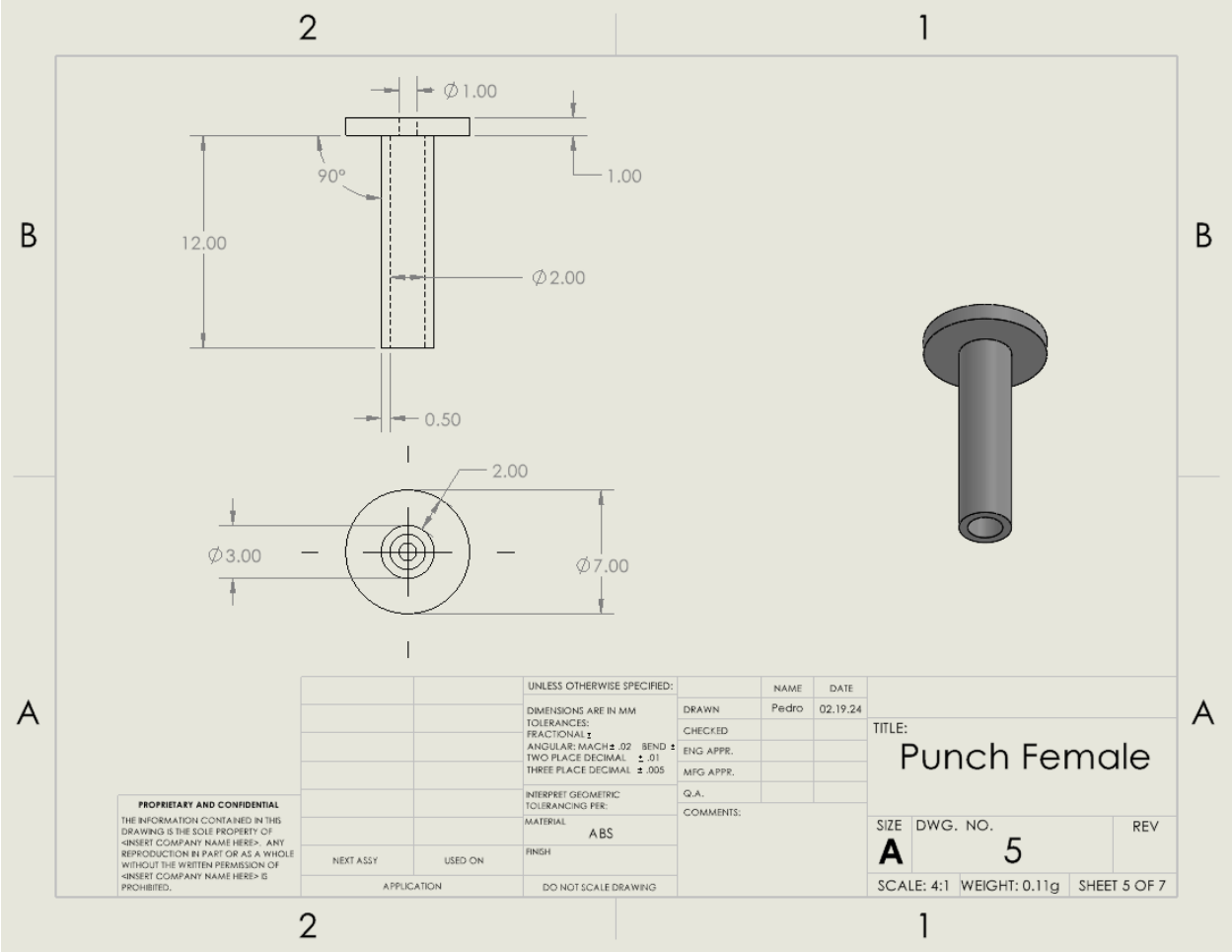


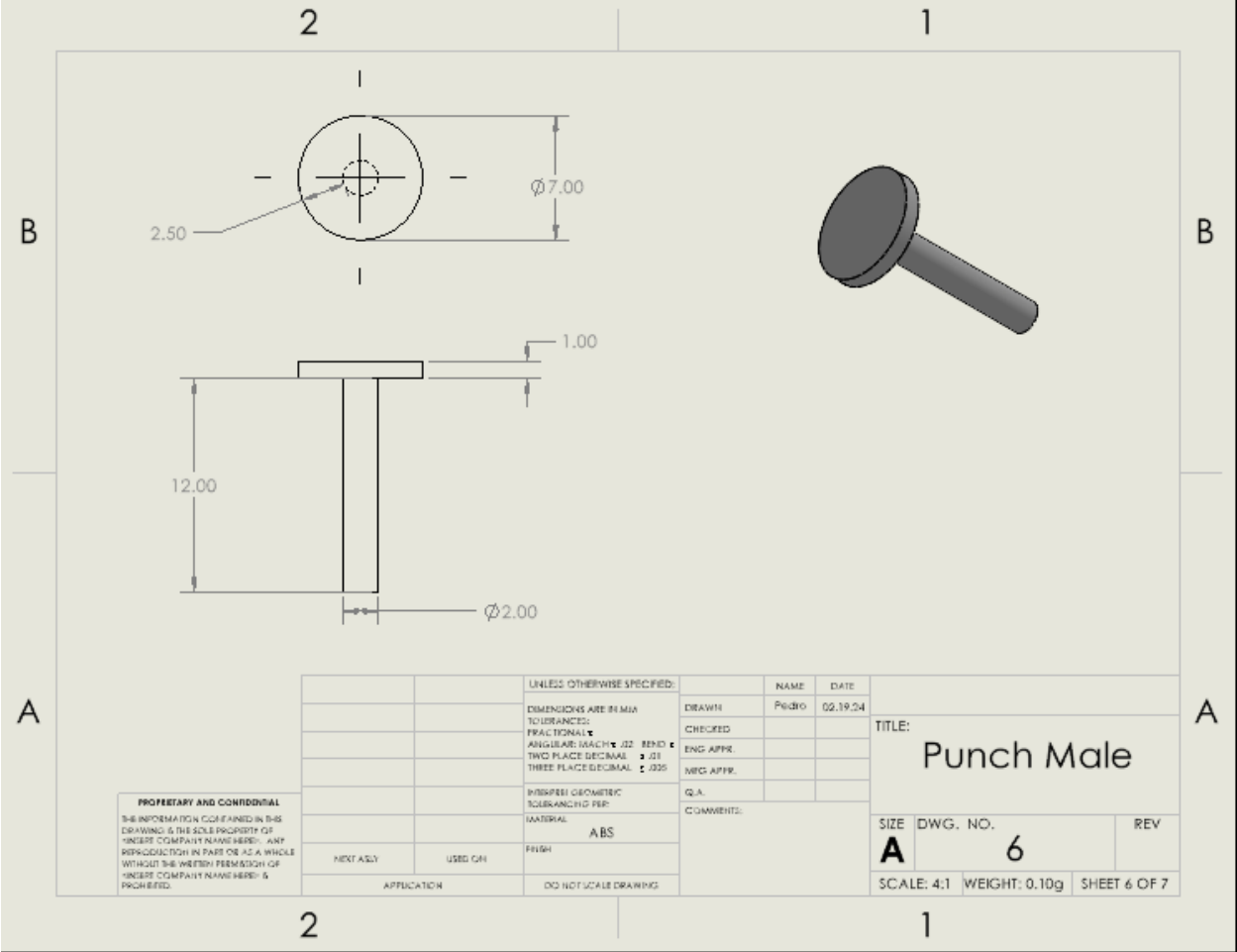


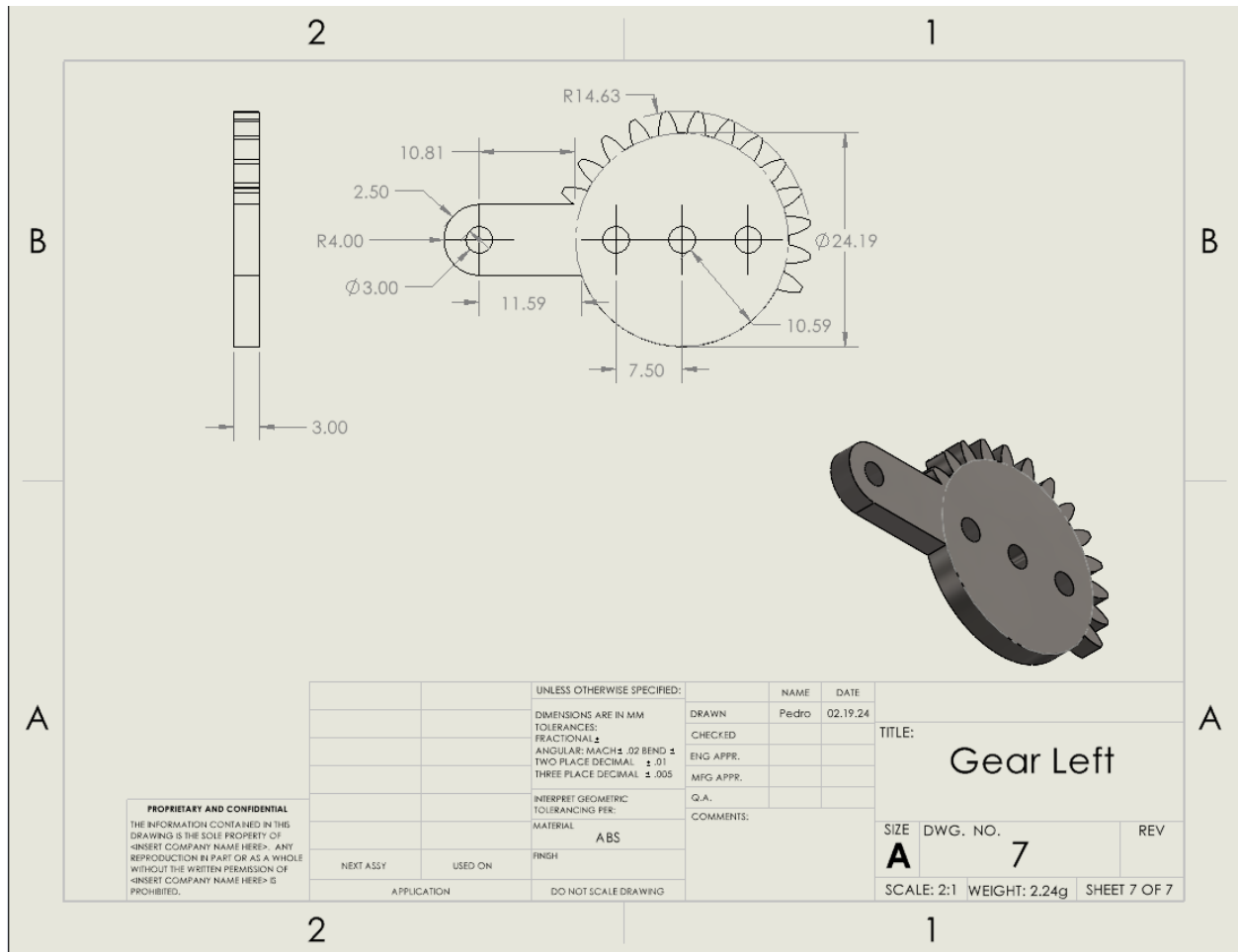












Torque Calculations

$$\gamma = L * a$$

$$F_{req} = \frac{m(2.3g + \gamma)}{\mu} FS$$

$$F_1 = F_{req} \cos \alpha$$

$$F_2 = \frac{F_1 \sin(\alpha + \beta)}{\sin \varphi}$$

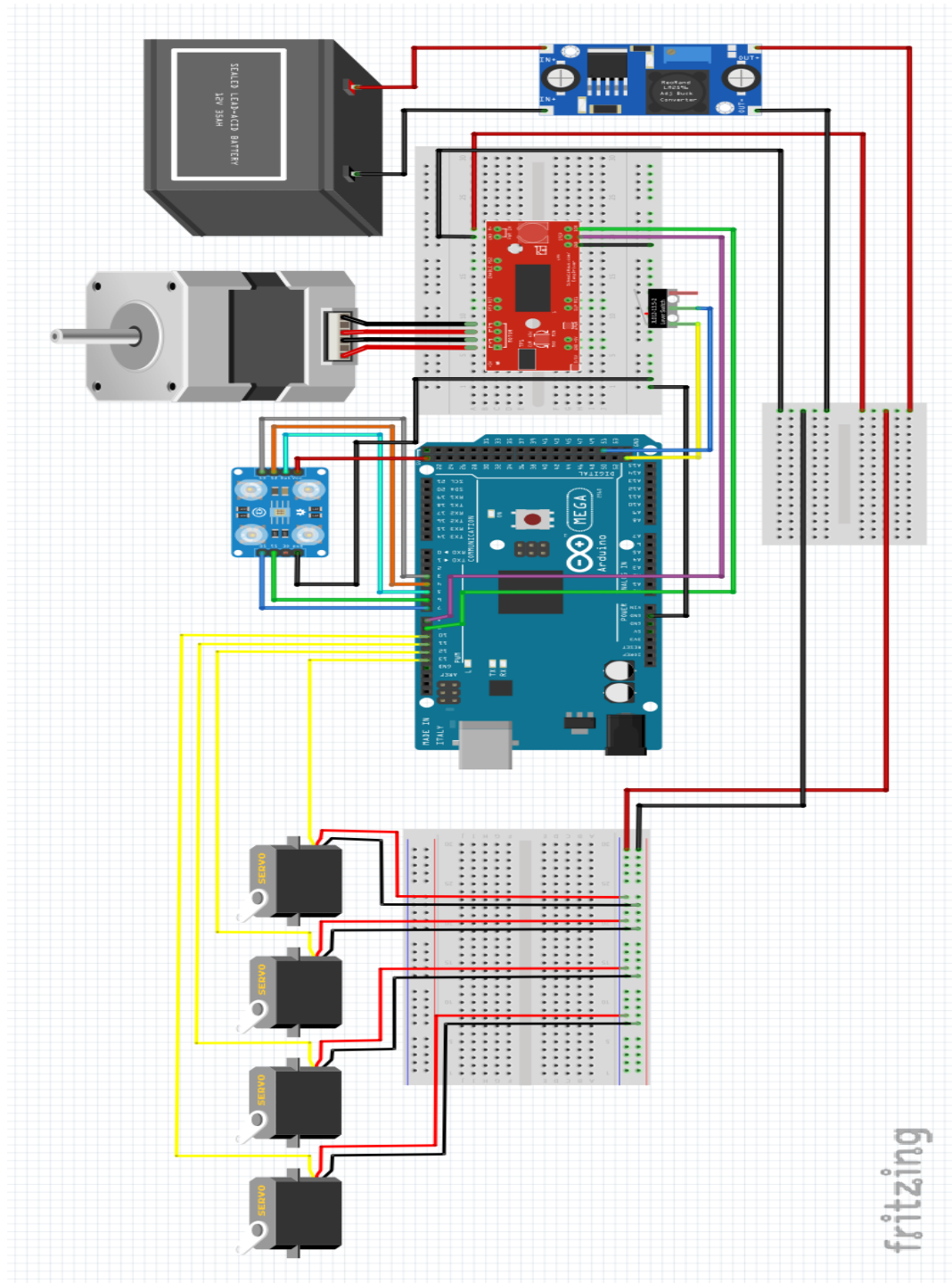
$$F_{gripper} = F_1 \cos(\alpha + \beta) + F_2 \cos \varphi$$

$$T_{req} = F_{gripper} * r$$

Variable (Inputs)	Definition	Value
L	Maximum Length of Robot	.3722m
a	Angular Acceleration (Robot)	$52.4 \frac{rads}{s^2}$
γ	Angular Acceleration (Cube)	$19.503 \frac{m}{s^2}$
m	Maass of cube	10g
g	Gravity Constant	$9.81 \frac{m}{s^2}$
μ	Coefficient of Friction	.63
FS	Factor of Safety	4
F_{req}	Force Required	2.6722
r	Radius of Rotation	.023m
Variable (Inputs @ minimum aperture)	Definition	Value
α	Angle	60.46°
β	Angle	7.54°
φ	Angle	82.46°
F_1	Force on Gear to Link joint	1.317N
F_2	Force on Link to Effector joint	1.233N
$F_{gripper}$	Force normal to Effector	.6550N
Variable (Inputs @ maximum)	Definition	Value
α	Angle	22.05°
β	Angle	67.95°
φ	Angle	0°
F_1	Force on Gear to Link joint	2.4767
F_2	Force on Link to Effector joint	0N
$F_{gripper}$	Force normal to Effector	2.4767N
Variable (Output)	Definition	Value
x	Total Required Torque (minimum aperture)	.01507Nm
T_{req}	Total Required Torque (maximum aperture)	.0569Nm

Controls

Wire Diagram



Components List

Component	Qty.
-----------	------

Arduino Mega 2560 Rev3	1
TCS34275 Color Sensor	1
Nema 17 Stepper Motor	1
12V Power Supply	1
LM2596 Buck Converter	1
SparkFun EasyDriver Stepper Motor Driver	1
RDS3235 35kg Servo Motor	1
RDS3225 25kg Servo Motor	1
LDX-218 17kg Servo Motor	1
USB Type-B Cable	1
Connecting Wires	≈ 40

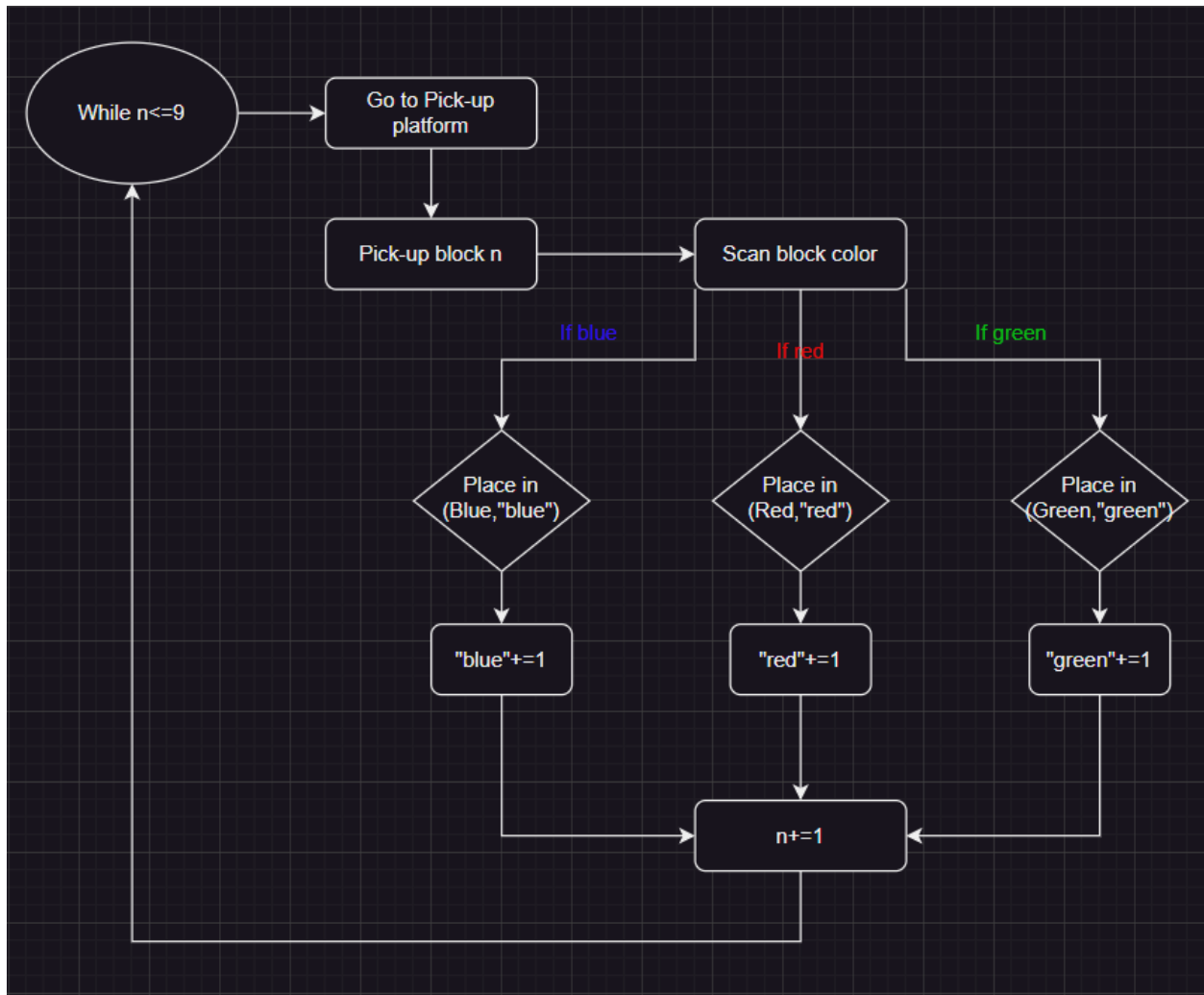
Power Estimates

$$P = VI = \frac{V^2}{R} = I^2 R$$

Components	Final Power
Nema 17 Stepper Motor	18W
TCS34275 Color Sensor	1.65mW
RA1 Servo Motor	13.3W
RA2	20.3W
Gripper Servo Motors	14.7W
Stepper Motor Driver	12.6W

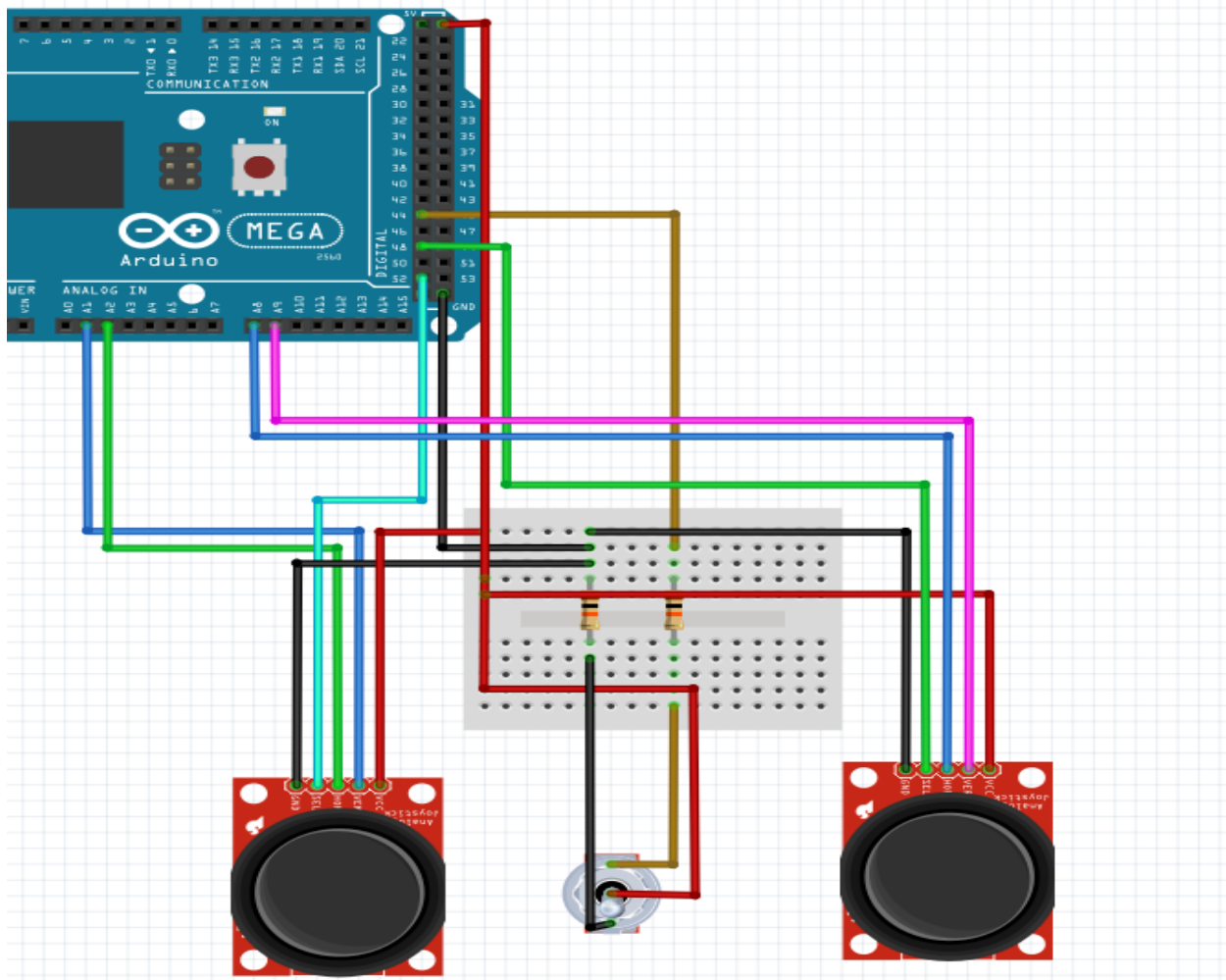
	Total
Amperage	11.9003A
Power	78.90165W

Code Flowchart

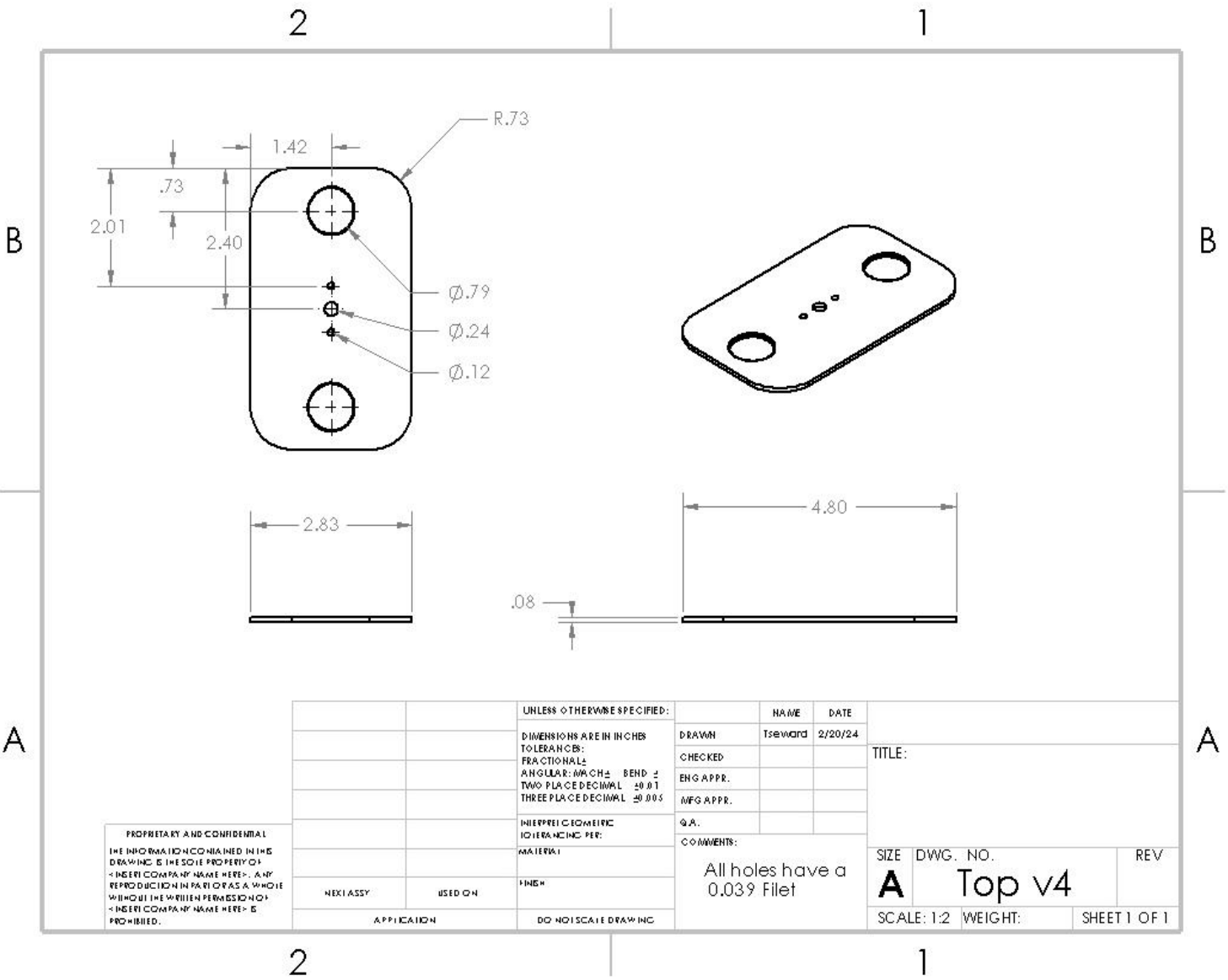


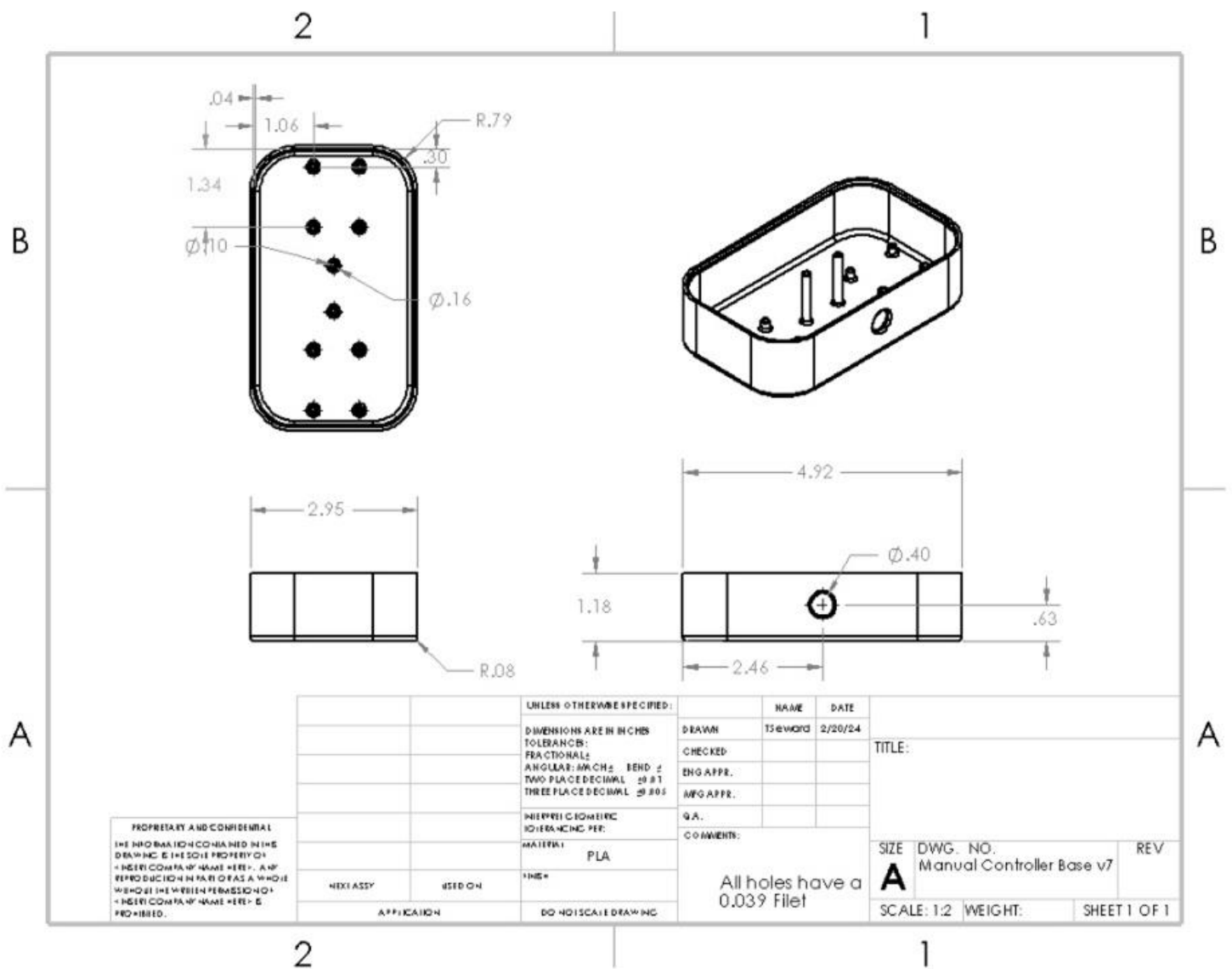
Manual Controls

Wire Diagram

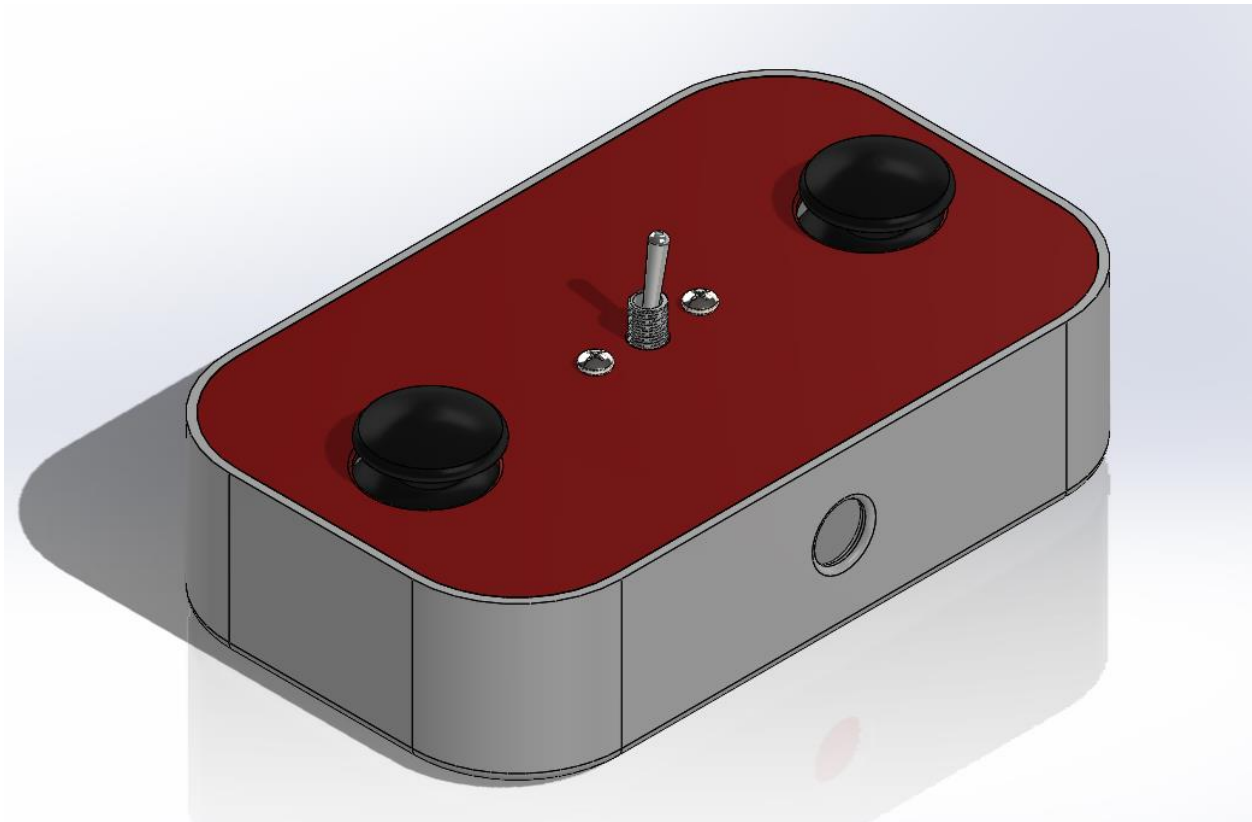


Drawings





Isometric View

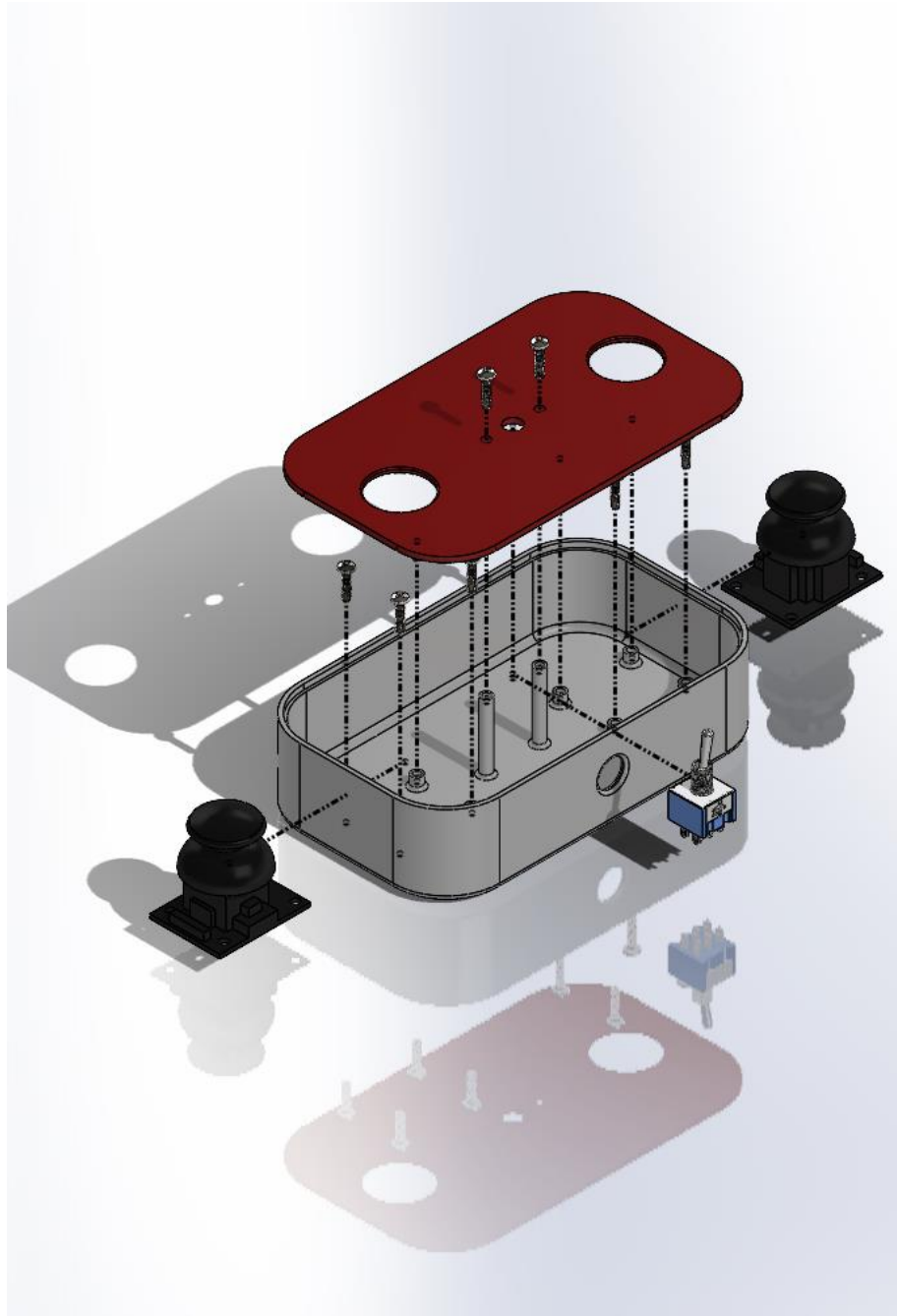


Components

Components	Qty.
Analog 2-Axis Thumb Joysticks	2
Toggle Switch	1
Controller Base	1
Controller Lid	1
Half Breadboard	1
Connection Wires	≈ 16
M3 Screws	10
10 k Ω Resistor	2

Power Estimates

Exploded View



Budget

Subsystem	Component	Material/Part Cost	Qty	Labor Cost	Total	Subsystem Total
Base	Custom Machined Base	\$ 26.11	1	\$ 20.00	\$ 46.11	\$ 46.35
	M3 Screws	\$ 0.02	4		\$ 0.08	
	M4 Screws	\$ 0.02	4		\$ 0.08	
	M5 Screws	\$ 0.02	4		\$ 0.08	
Linear Axis	Nema 17	\$ 8.99	1		\$ 8.99	\$ 101.36
	Linear Motion Shaft Kit	\$ 31.99	1		\$ 31.99	
	Base Support	\$ 10.55	1		\$ 10.55	
	Lead Screw Assembly	\$ 46.88	1		\$ 46.88	
	Microswitch	\$ 2.95	1		\$ 2.95	
RA 1	Servo Motor Assembly	\$ 36.59	1		\$ 36.59	\$ 43.43
	3D Printed Link	\$ 6.84	1		\$ 6.84	
RA 2	Servo Motor Assembly	\$ 21.99	1		\$ 21.99	\$ 23.02
	3D Printed Link	\$ 1.03	1		\$ 1.03	
Gripper	Actuation Servo	\$ 18.99	1		\$ 18.99	\$ 34.81
	Color Sensor	\$ 6.99	1		\$ 6.99	
	Gripper Servo	\$ 7.39	1		\$ 7.39	
	Heat Set Inserts	\$ 0.06	4		\$ 0.24	
	3D Printed Gripper	\$ 1.20	1		\$ 1.20	
Controls	Arduino	\$ 51.91	1		\$ 51.91	\$ 127.27
	Pack of Wires	\$ 12.00	1		\$ 12.00	
	Toggle Switch	\$ 8.00	1		\$ 8.00	
	3D Printed Controller	\$ 15.70	1		\$ 15.70	
	Joysticks [2 pack]	\$ 13.00	1		\$ 13.00	
	Buck Converter	\$ 5.49	1		\$ 5.49	
	Stepper Motor Driver	\$ 18.17	1		\$ 18.17	
	Zip Ties	\$ 3.00	1		\$ 3.00	
Materials Subtotal						\$ 376.24
Student Hours	Design Hours		180	\$ 10.00	\$ 1,800.00	
	Assembly Hours		120	\$ 10.00	\$ 1,200.00	
	Testing Hours		75	\$ 10.00	\$ 750.00	
Labor Subtotal						\$ 3,750.00
Robot Total						\$ 4,126.24

Kinematics and Uncertainty

$$A(x) = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$A(y) = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

Where:

L_1	15.5 cm
L_2	13 cm
L_3	8 cm

Inverse Kinematics

$$C_{2x} = x - L_3 \cos(\alpha)$$

$$C_{2y} = y - L_3 \sin(\alpha)$$

$$\theta_2 = \arccos\left(\frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}\right)$$

$$\theta_1 = \arctan\left(\frac{y}{x}\right) - \arctan\left(\frac{L_2 \sin(\theta_2)}{L_1 + L_2 \cos(\theta_2)}\right)$$

α	$0 = \theta_1 + \theta_2 + \theta_3$
----------	--------------------------------------

Uncertainty

$$u_c^2(x) = \left(\frac{\partial x}{\partial L_1}\right)^2 u^2(L_1) + \left(\frac{\partial x}{\partial L_2}\right)^2 u^2(L_2) + \left(\frac{\partial x}{\partial L_3}\right)^2 u^2(L_3) + \left(\frac{\partial x}{\partial \theta_1}\right)^2 u^2(\theta_1) + \left(\frac{\partial x}{\partial \theta_2}\right)^2 u^2(\theta_2) + \left(\frac{\partial x}{\partial \theta_3}\right)^2 u^2(\theta_3)$$

$$u_c^2(y) = \left(\frac{\partial y}{\partial L_1}\right)^2 u^2(L_1) + \left(\frac{\partial y}{\partial L_2}\right)^2 u^2(L_2) + \left(\frac{\partial y}{\partial L_3}\right)^2 u^2(L_3) + \left(\frac{\partial y}{\partial \theta_1}\right)^2 u^2(\theta_1) + \left(\frac{\partial y}{\partial \theta_2}\right)^2 u^2(\theta_2) + \left(\frac{\partial y}{\partial \theta_3}\right)^2 u^2(\theta_3)$$

$$\left(\frac{\partial x}{\partial L_1}\right) = \cos(\theta_1)$$

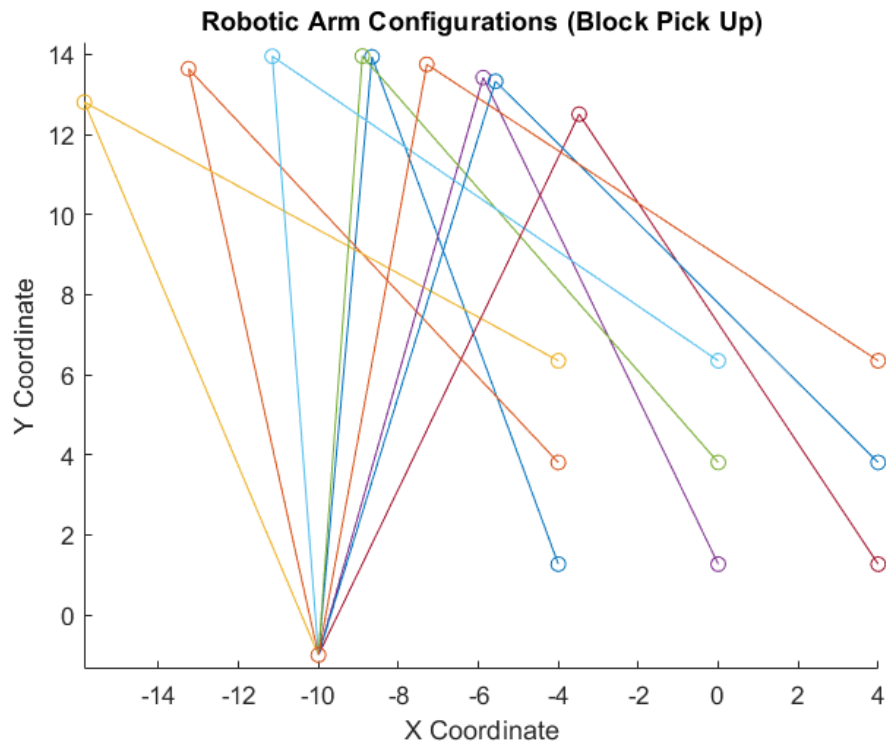
$$\left(\frac{\partial x}{\partial L_2}\right) = \cos(\theta_1 + \theta_2)$$

$$\left(\frac{\partial x}{\partial L_3}\right) = \cos(\theta_1 + \theta_2 + \theta_3)$$

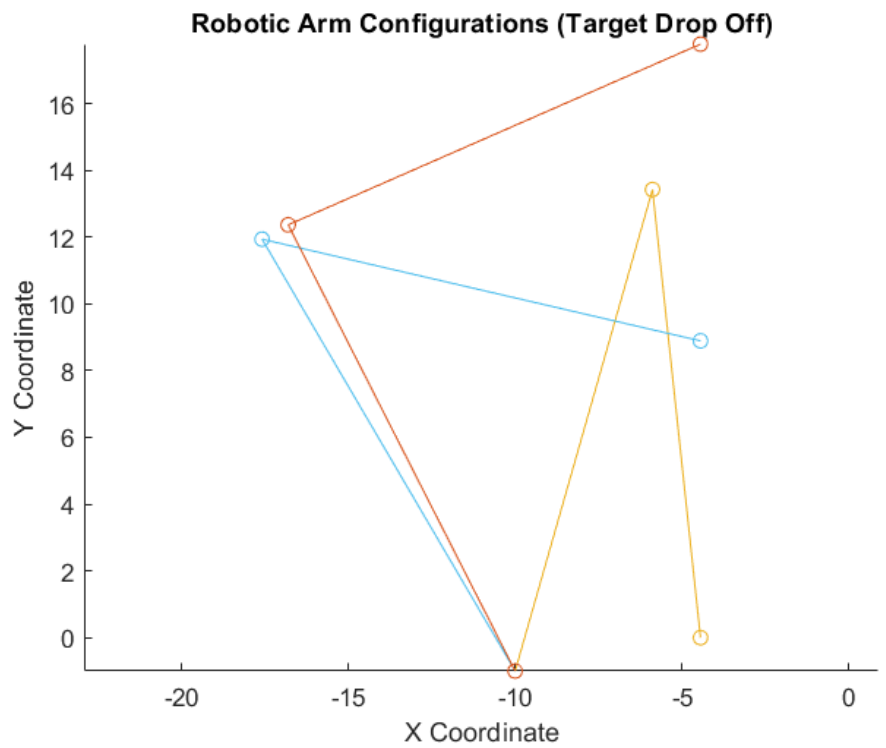
Where

u(all link lengths)	.56 mm
u(all angles)	0.5°

Kinematic Results for Block Pick Up



Kinematic Results for Block Drop Off



Uncertainty Results

Targets/Blocks	95% Confidence X	95% Confidence Y
T4	5.540	3.802
T7	5.280	4.176
B3	5.078	4.425
B9	4.736	4.818
B6	4.642	4.892
B8	4.512	5.077
B7	4.386	5.239
B2	4.350	5.199
B5	4.196	5.336
B4	3.890	5.629
B1	3.561	5.850
T1	3.519	5.916

